50 published mechanical FEM problems

Original source drawings, geometry illustrations and numerical FEM baselines

This release contains 50 distinct physical problems from EDF's official Code_Aster validation library. Each has a publisher geometry illustration, an original solver command file, its declared input files, and numerical FEM baseline assertions. No engineering problem, CAD geometry or FEM reference was generated here.

Source release - benchmark admission pending

The illustrations include technical geometry sketches and some model views. They are not all native CAD or manufacturing drawings. No STEP files were found in this release. Original GIBI geometry definitions are included where the publisher provides them; MED files are FE meshes. These distinctions are recorded per case.

The numerical values shown are published VALE_CALC FEM baselines, not model benchmark scores. They remain in original source units until unit and output-selector review. Source comparison tolerances are not adopted as model acceptance tolerances. The executable pilot still admits 3 cases; this release adds 50 source pairs for review and does not claim 50 certified scored cases.

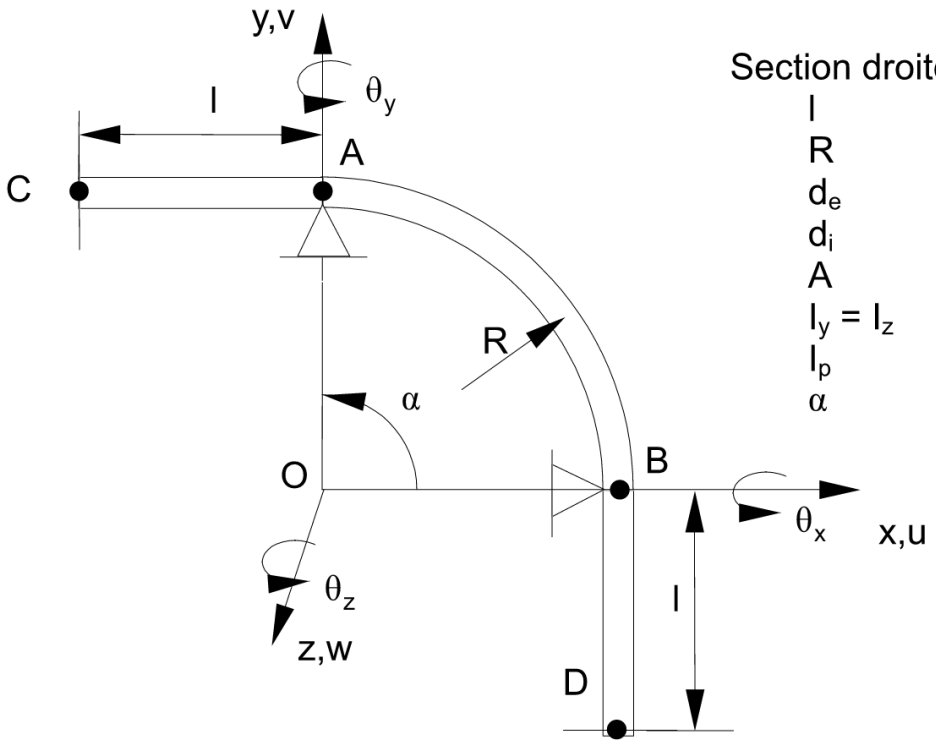
Physics family	Problems
vibration and rotor dynamics	9
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fracture and crack propagation	9
contact and friction	8
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thermomechanics	3

Evaluator material: the source decks and this PDF contain answers. Do not expose them to candidate models. The two benchmark tracks require separate answer-free input preparation.

01 / 50 ASTER-SDLL14

Vibration modes of a thin pipe elbow

vibration and rotor dynamics



Section droite circulaire creuse

l	=	2.0 m
R	=	1. m
d_e	=	0.020 m
d_i	=	0.016 m
A	=	1.131×10^{-4} m²
$I_y = I_z$	=	4.637×10^{-9} m^4
I_p	=	9.274×10^{-9} m^4
α	=	90°

Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NUME_ORDRE=1; PARA=FREQ	17.54133946	135
NUME_ORDRE=2; PARA=FREQ	24.28045138	145
NUME_ORDRE=3; PARA=FREQ	24.78831794	155

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

Original input: sdll14a.comm. Geometry definition: publisher diagram and original FE mesh; no separate CAD definition verified.

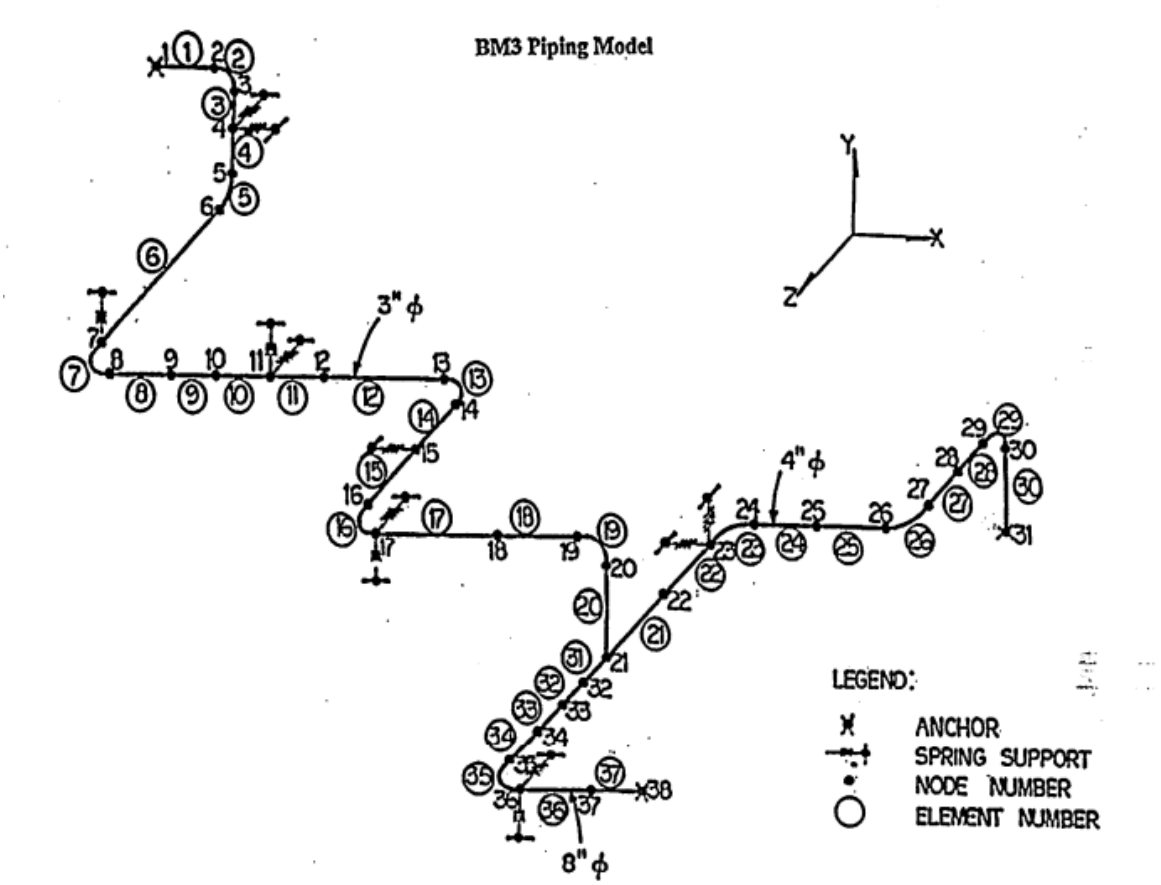
Publisher documentation: SDLL14

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

02 / 50 ASTER-SDLL149

Seismic response of the NRC BM3 piping system

vibration and rotor dynamics



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
PARA=FREQ	2.904748208	248
PARA=FREQ	4.393138077	257
PARA=FREQ	4.833952846	266

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

Original input: sdll149a.comm. Geometry definition: original GIBI script supplied.

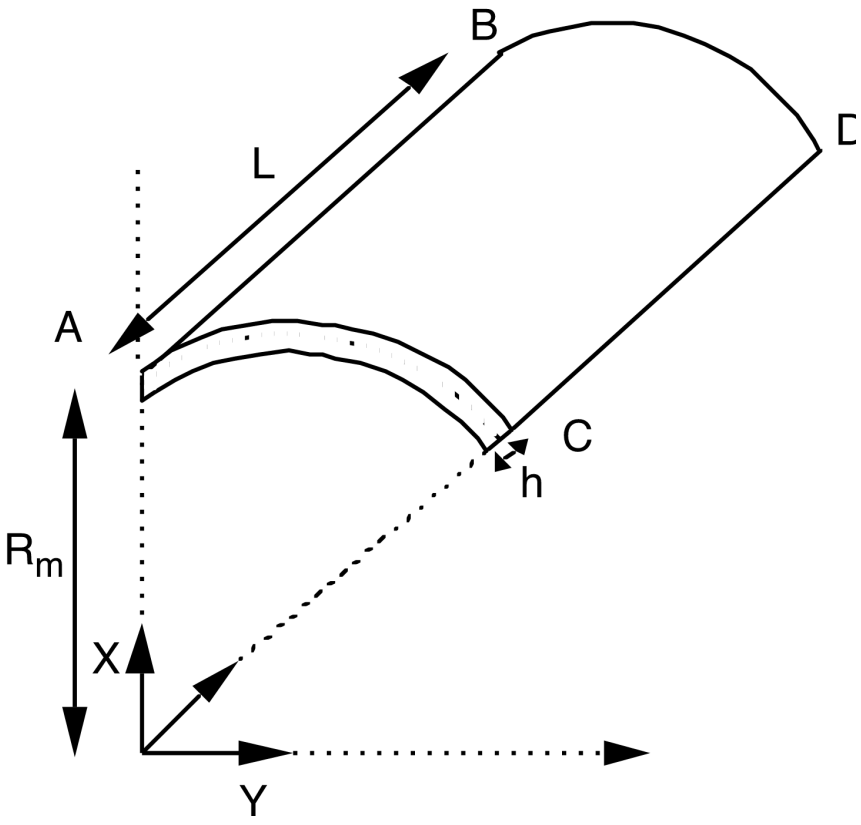
Publisher documentation: SDLL149

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

03 / 50 ASTER-SDLS102

Natural frequencies of a compressor blade

vibration and rotor dynamics



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NUME_ORDRE=1; PARA=FREQ	85.85330632	72
NUME_ORDRE=2; PARA=FREQ	138.5571264	82
NUME_ORDRE=3; PARA=FREQ	246.8731624	92

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

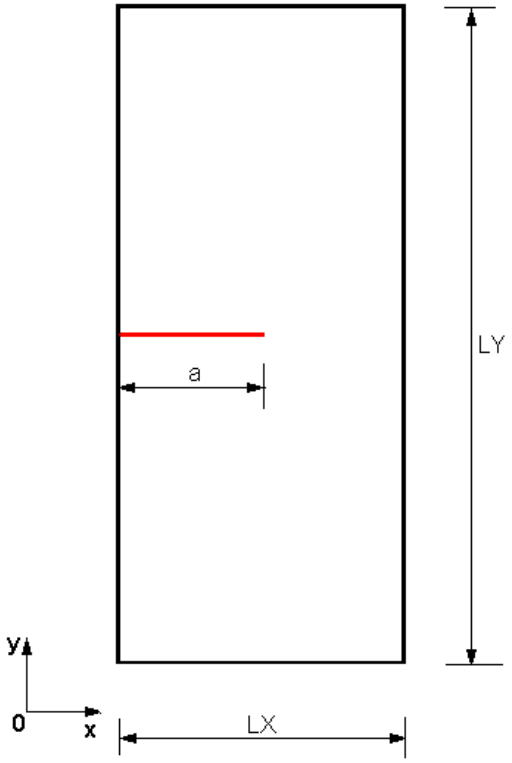
Original input: sdls102a.comm. Geometry definition: publisher diagram and original FE mesh; no separate CAD definition verified.

Publisher documentation: SDLS102

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Vibration modes of a cracked plate with X-FEM

vibration and rotor dynamics



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NUME_ORDRE=1; PARA=FREQ	7.005811696	153
NUME_ORDRE=2; PARA=FREQ	24.89548844	159
NUME_ORDRE=3; PARA=FREQ	41.82065759	165

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

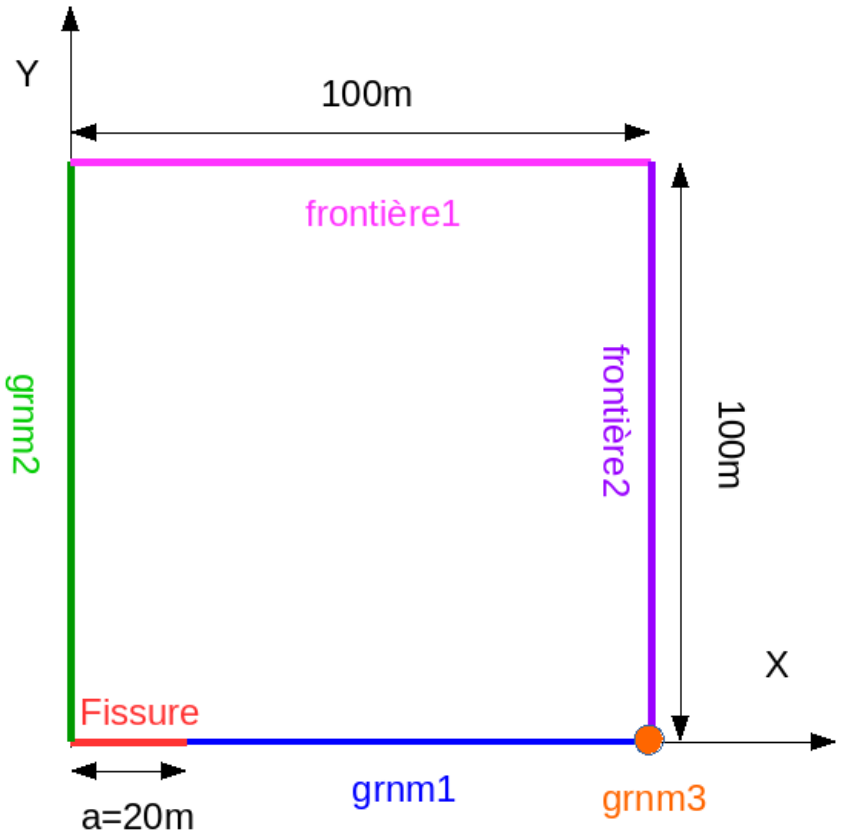
Original input: sdls120a.comm. Geometry definition: original GIBI script supplied.

Publisher documentation: SDLS120

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Elastodynamic energy release rate of a finite crack

vibration and rotor dynamics



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_PARA=G_EPSI	564807.0083	170
NOM_PARA=G_EPSI	1129734.348	180
NOM_PARA=G_EPSI	1683121.425	190

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

Original input: sdlv123a.comm. Geometry definition: original GIBI script supplied.

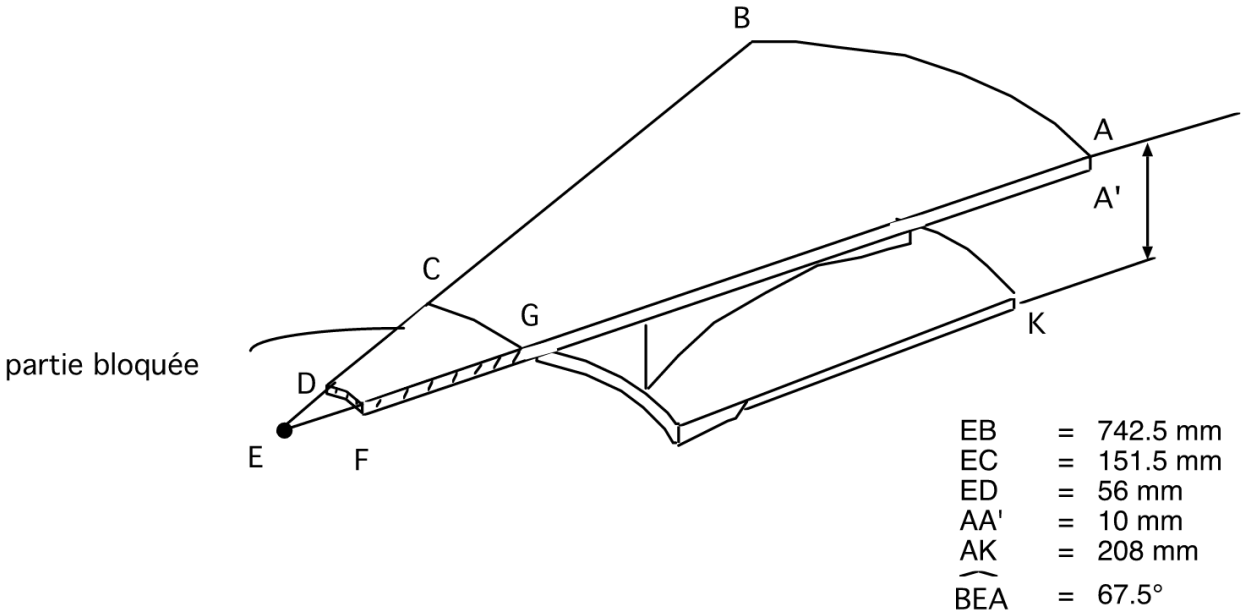
Publisher documentation: SDLV123

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

06 / 50 ASTER-SDLV129

Vibration fatigue of a fan blade

vibration and rotor dynamics



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_CHAM=SIEQ_ELNO; NOM_CMP=VMIS_SG; GROUP_NO=N111; GROUP_MA=M294; NUME_ORDRE=1	307.7125149	64
NOM_CHAM=SIEQ_ELNO; NOM_CMP=VMIS; GROUP_NO=N111; GROUP_MA=M294; NUME_ORDRE=1	307.7125149	76
NOM_CHAM=SIEQ_ELNO; NOM_CMP=VMIS_SG; GROUP_NO=N111; GROUP_MA=M294	9.800254947	114

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

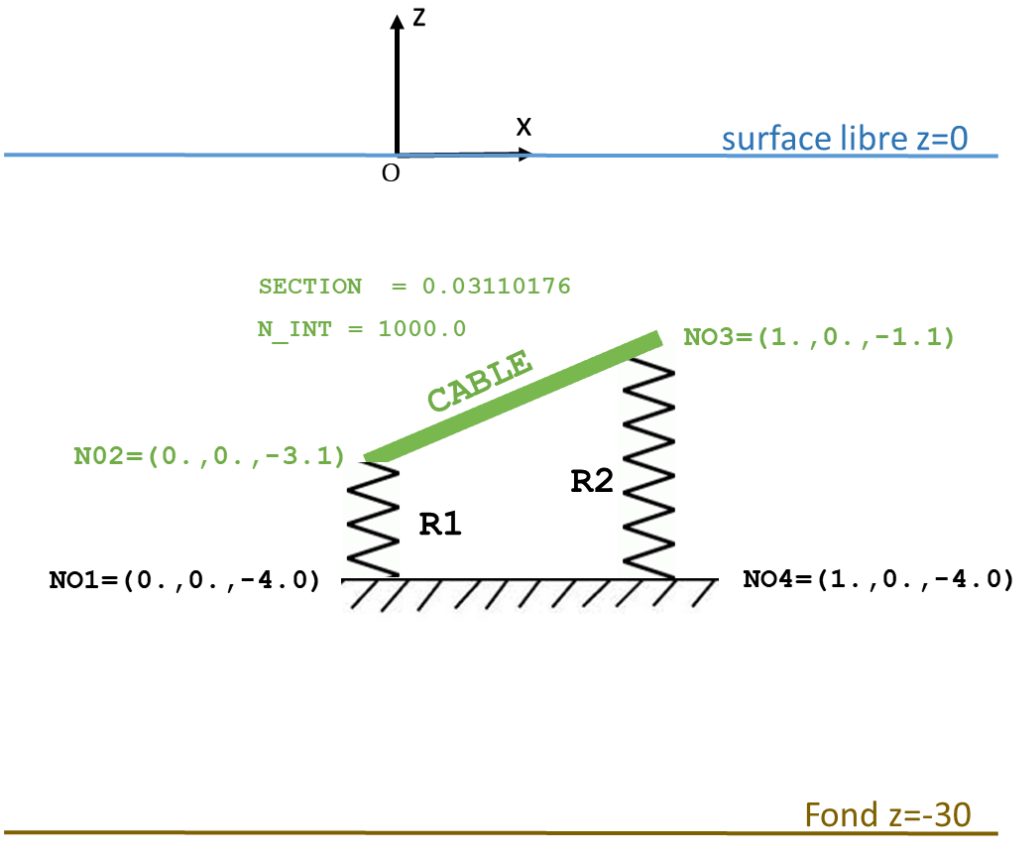
Original input: sdlv129a.comm. Geometry definition: publisher diagram and original FE mesh; no separate CAD definition verified.

Publisher documentation: SDLV129

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Dynamic response of a submerged cable in a wave field

vibration and rotor dynamics



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_CHAM=DEPL; NOM_CMP=DX; GROUP_NO=N0002; INST=1.0	0.3190649621	188
NOM_CHAM=DEPL; NOM_CMP=DZ; GROUP_NO=N0002; INST=1.0	-0.1595324811	198
NOM_CHAM=DEPL; NOM_CMP=DX; GROUP_NO=N0002; INST=2.0	0.8361208713	210

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

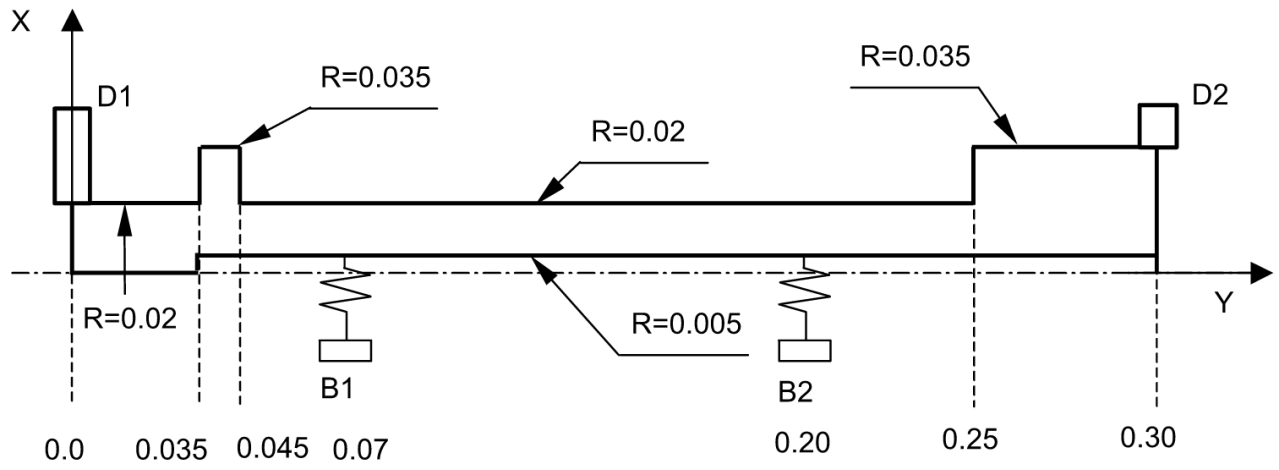
Original input: sdnl142a.comm. Geometry definition: publisher diagram and original FE mesh; no separate CAD definition verified.

Publisher documentation: SDNL142

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Gyroscopic harmonic response of a two-disc rotor

vibration and rotor dynamics



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_PARA=DEPL	1.03312867e-07	371
NOM_PARA=DEPL	1.045701767e-07	381
NOM_PARA=DEPL	1.938083883e-07	391

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

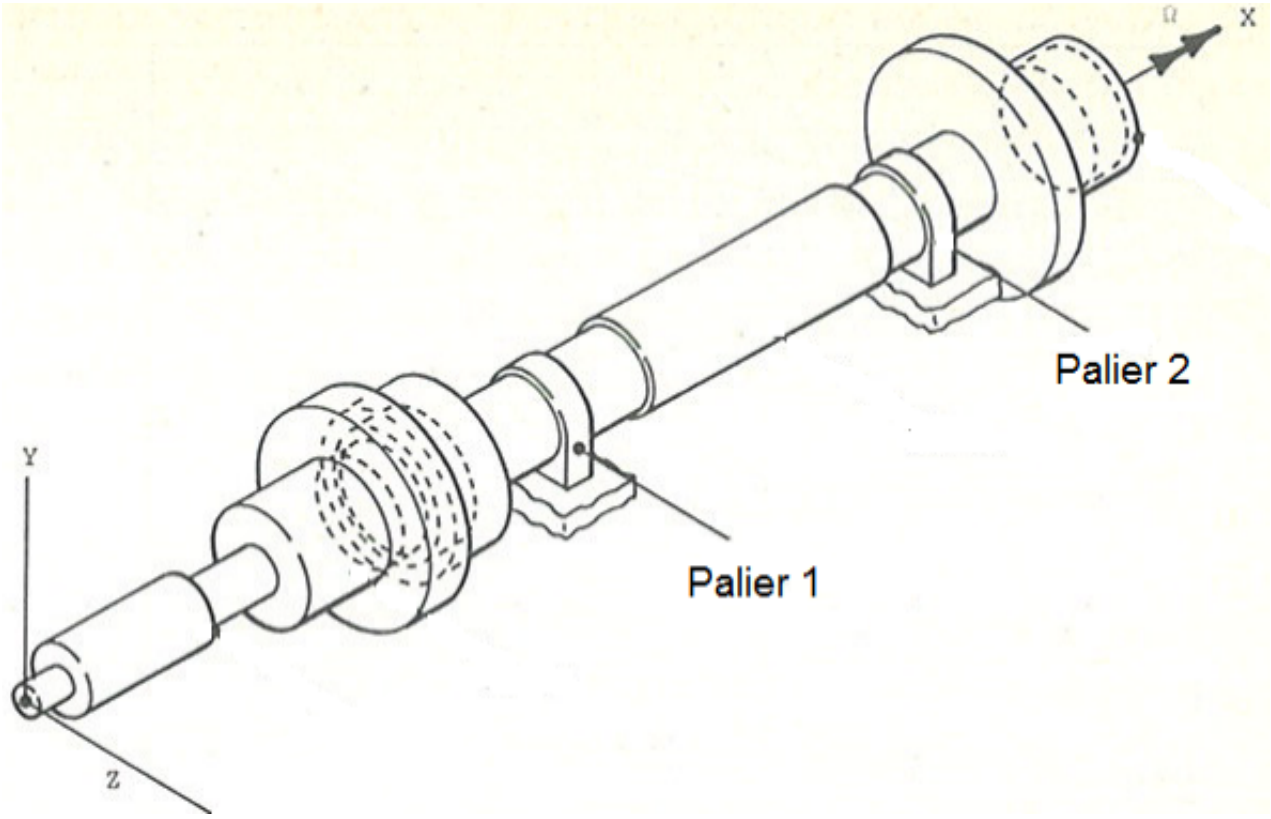
Original input: shll103a.comm. Geometry definition: publisher diagram and original FE mesh; no separate CAD definition verified.

Publisher documentation: SHLL103

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Gyroscopic modes of a rotating solid rotor

vibration and rotor dynamics



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NUME_ORDRE=1; PARA=FREQ	55.95132468	156
NUME_ORDRE=2; PARA=FREQ	176.3207331	166
NUME_ORDRE=3; PARA=FREQ	501.4259798	176

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

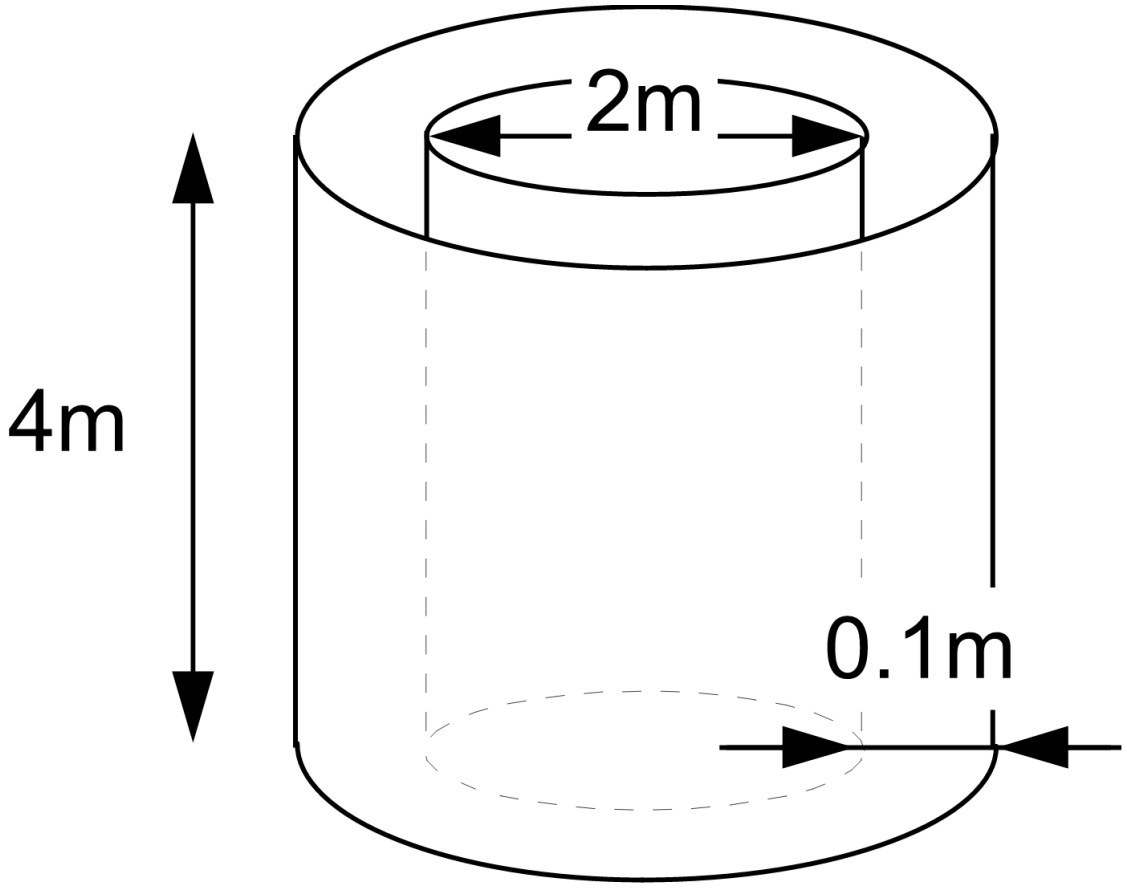
Original input: sdnv110a.comm. Geometry definition: publisher diagram and original FE mesh; no separate CAD definition verified.

Publisher documentation: SDNV110

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Inertial coupling of coaxial shells in annular flow

fluid-structure coupling



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
Original TEST_RESU / TEST_TABLE selector	5.295950407	134
Original TEST_RESU / TEST_TABLE selector	5.296253752	151
Original TEST_RESU / TEST_TABLE selector	6.121135803	168

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

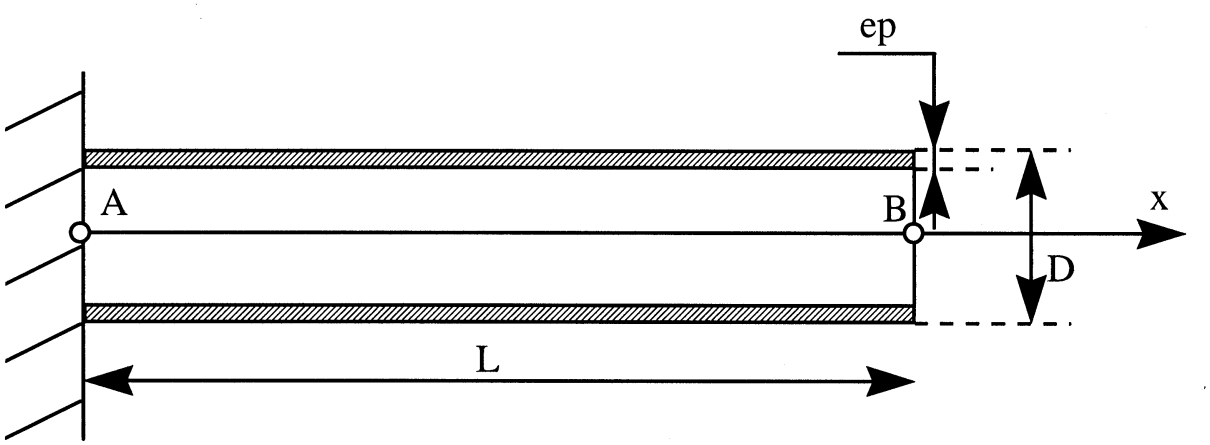
Original input: sdls103a.comm. Geometry definition: publisher diagram and original FE mesh; no separate CAD definition verified.

Publisher documentation: SDLS103

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Vibration of a fluid-filled cantilever pipe

fluid-structure coupling



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NUME_ORDRE=1; PARA=FREQ	157.9453912	93

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

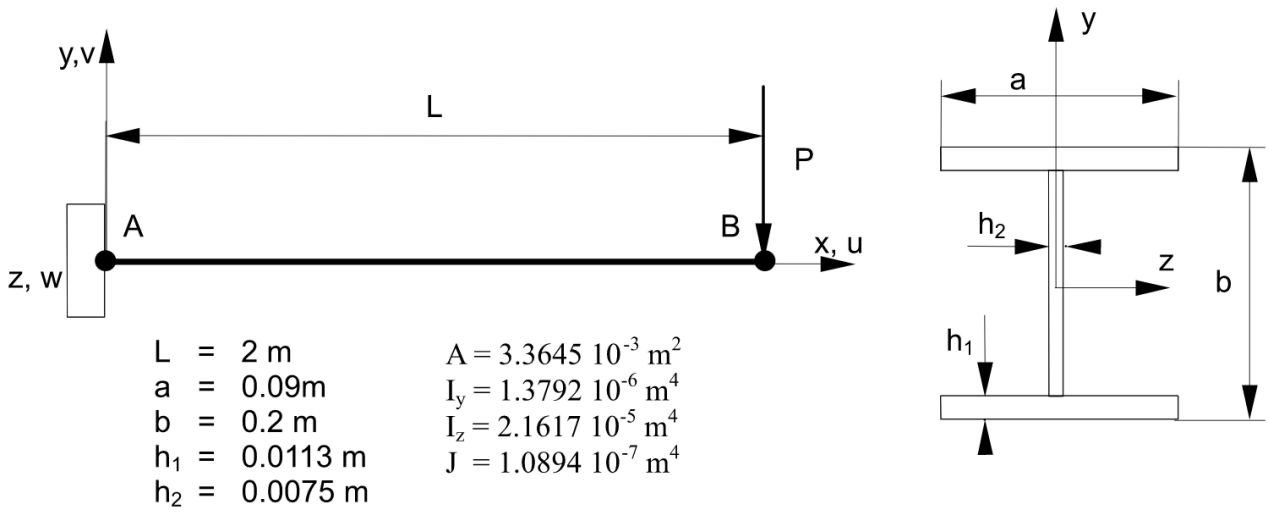
Original input: fdll200a.comm. Geometry definition: publisher diagram and original FE mesh; no separate CAD definition verified.

Publisher documentation: FDLL200

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Lateral torsional buckling of a beam

buckling and stability



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
PARA=CHAR_CRIT	-107753.483	212
PARA=CHAR_CRIT	107878.765	222
PARA=CHAR_CRIT	-107753.4833	237

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

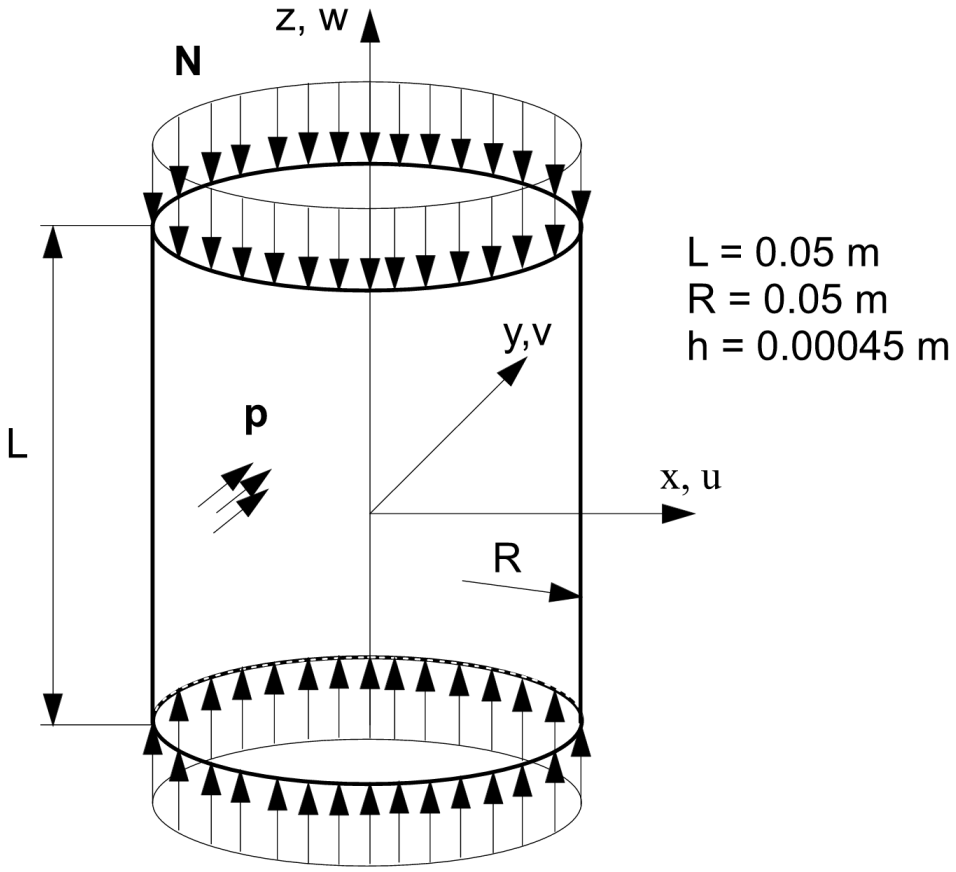
Original input: sdls504a.comm. Geometry definition: publisher diagram and original FE mesh; no separate CAD definition verified.

Publisher documentation: SDLS504

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Cylindrical shell buckling under external pressure

buckling and stability



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
PARA=CHAR_CRIT	1686182.546	129
NOM_CMP=DX; GROUP_NO=NO1008	-0.6445033317	172
NOM_CMP=DY; GROUP_NO=NO1008	0.2537488811	173

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

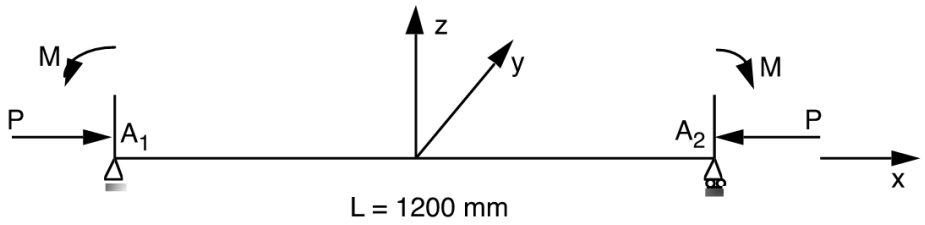
Original input: sdls505a.comm. Geometry definition: publisher diagram and original FE mesh; no separate CAD definition verified.

Publisher documentation: SDLS505

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Elastic buckling of an angle-section member

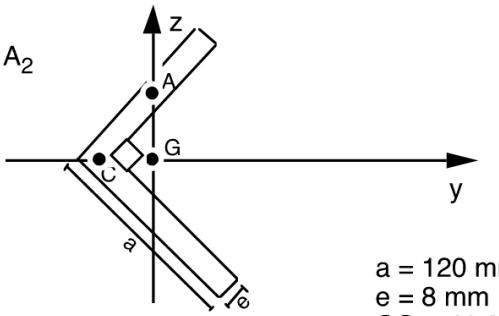
buckling and stability



Caractéristiques de la section

$A = 1856\ mm^2$
$I_y = 4167339\ mm^4$
$I_z = 1045547\ mm^4$
$J = 39595\ mm^4$
$I_\omega = 44398819\ mm^6$
$I_{yr2} = 84948392\ mm^5$
$z_c = 0$

Section A_1 et A_2



a = 120 mm
e = 8 mm
CG = 41.012 mm

Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
PARA=CHAR_CRIT	691154.293	185
PARA=CHAR_CRIT	1490047.9	194
PARA=CHAR_CRIT	9694294.78	203

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

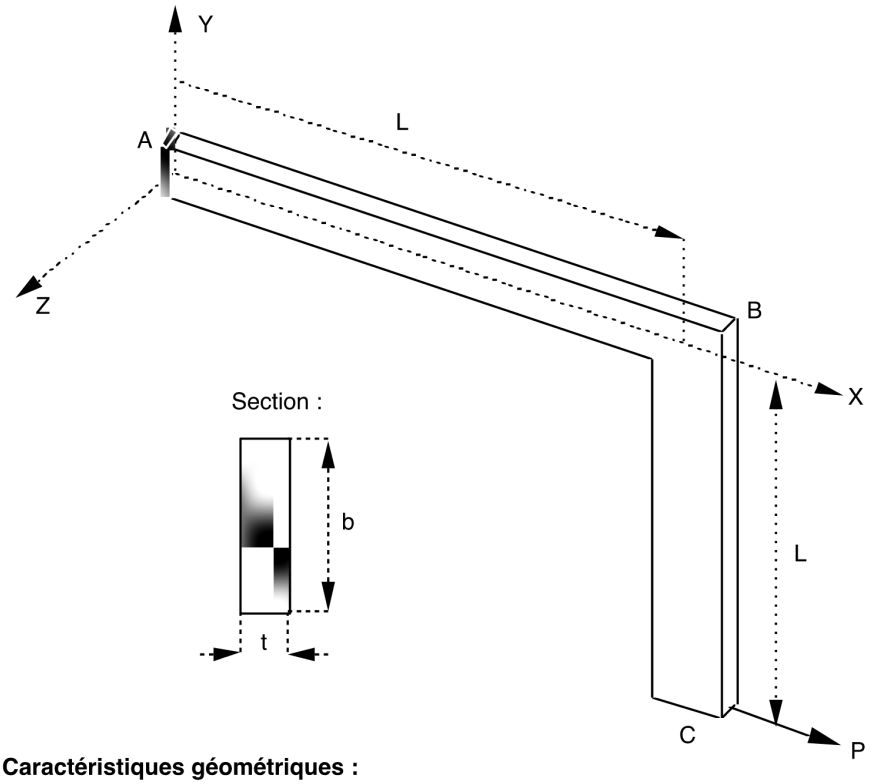
Original input: ssll103a.comm. Geometry definition: publisher diagram and original FE mesh; no separate CAD definition verified.

Publisher documentation: SSLL103

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Elastic buckling of an L-shaped structure

buckling and stability



Caractéristiques géométriques :

L = 240 mm
b = 30 mm
t = 0.6 mm

Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
PARA=CHAR_CRIT	-1.090064344	86
PARA=CHAR_CRIT	-0.681323581	126

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

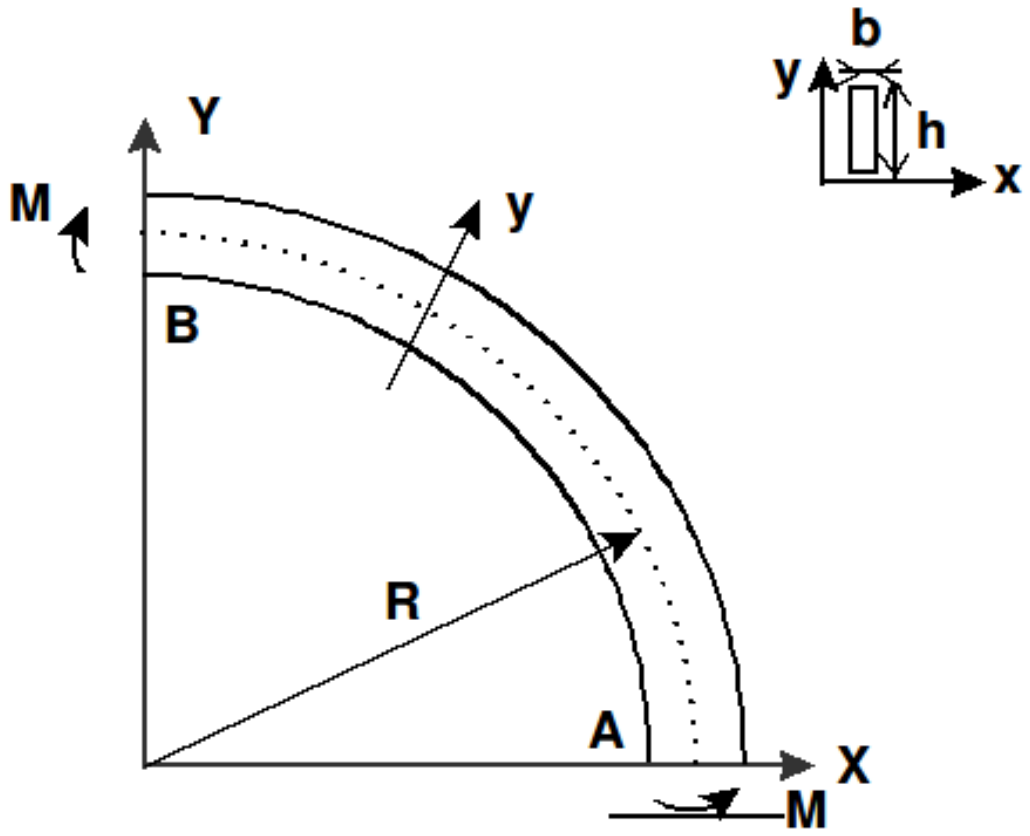
Original input: ssll105a.comm. Geometry definition: publisher diagram and original FE mesh; no separate CAD definition verified.

Publisher documentation: SSLL105
Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

16 / 50 ASTER-SSLL404

Buckling of a curved arch

buckling and stability



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
PARA=CHAR_CRIT	13.93216399	111
PARA=CHAR_CRIT	8.306132479	121
PARA=CHAR_CRIT	2.751372218	131

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

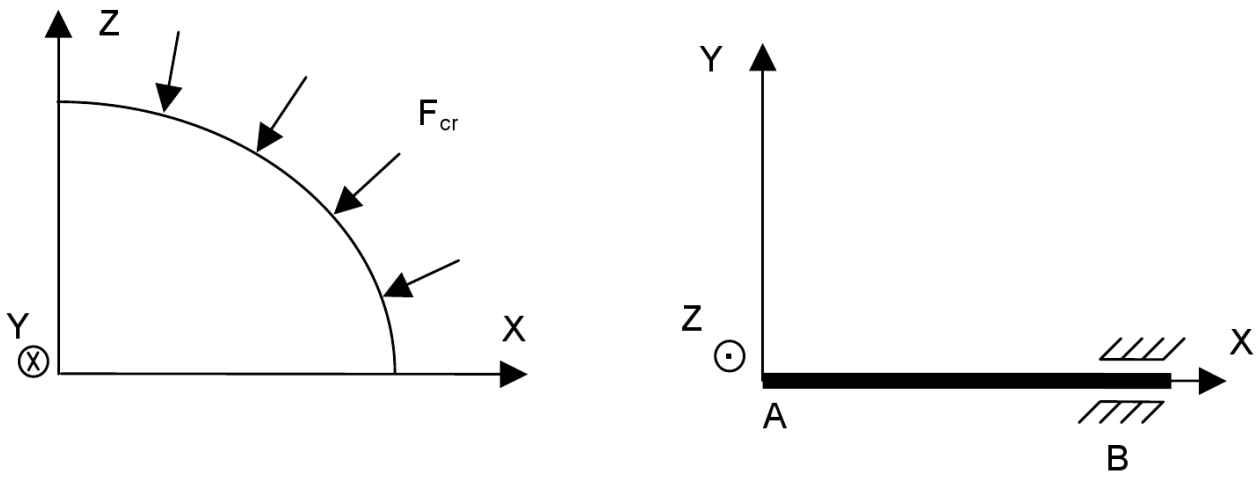
Original input: ssll404a.comm. Geometry definition: publisher diagram and original FE mesh; no separate CAD definition verified.

Publisher documentation: SSLL404

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Buckling of a circular plate under edge compression

buckling and stability



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
PARA=CHAR_CRIT	-848467.4365	105
NOM_CMP=DY; GROUP_NO=N4	0.9999854495	118

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

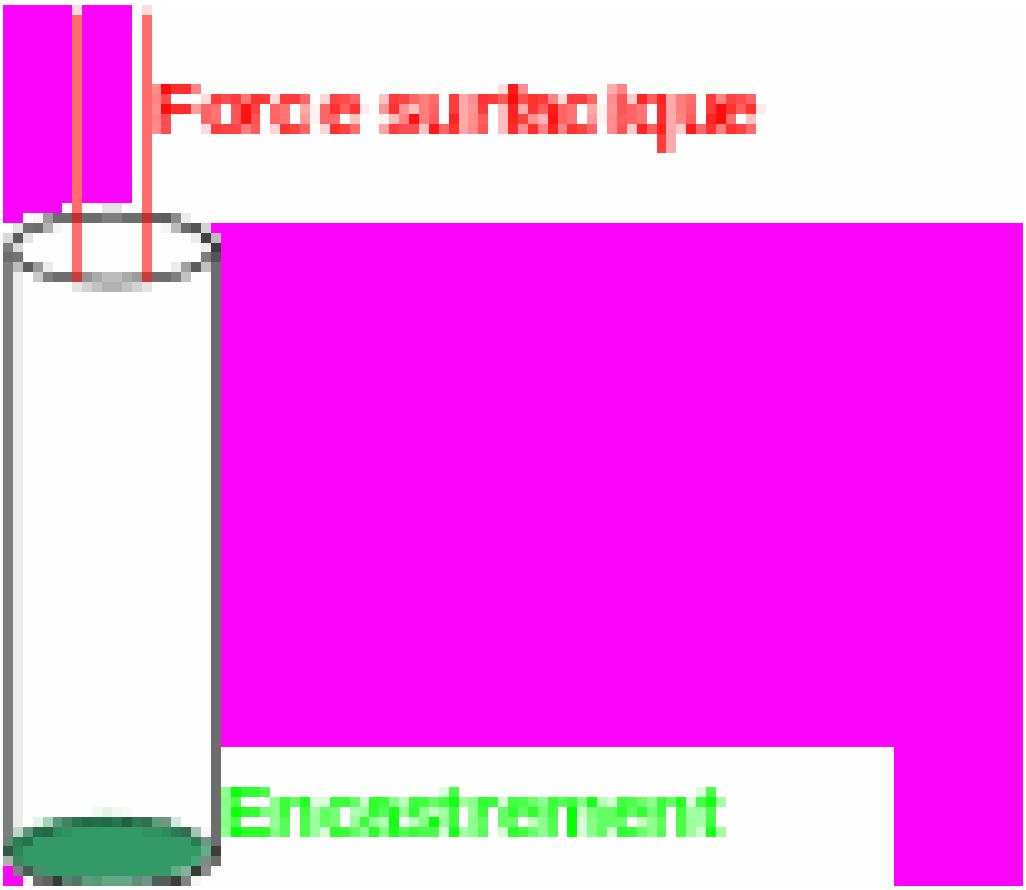
Original input: sslv139a.comm. Geometry definition: original GIBI script supplied.

Publisher documentation: SSLV139

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Elastoplastic buckling of a straight beam

buckling and stability



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_PARA=CHAR_CRIT	9.97636641	57
NOM_PARA=CHAR_CRIT	0.675266654	72

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

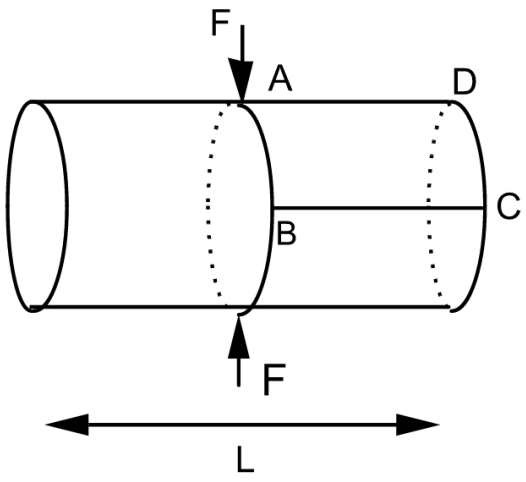
Original input: ssnl126a.comm. Geometry definition: original GIBI script supplied.

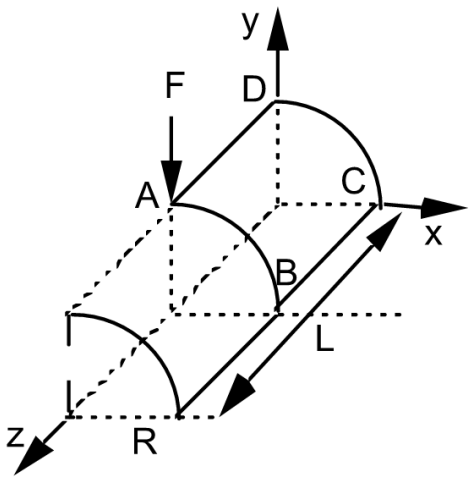
Publisher documentation: SSNL126

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Pinched cylindrical shell with free edges

shells and composites





huitième de cylindre

Longueur	L = 10.35 m
Rayon	R = 4.953 m
Epaisseur	t = 0.094 m

Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_CMP=DY; GROUP_NO=NO176	-0.11328222	78
NOM_CMP=DY; GROUP_NO=NO176	-0.111412003	114
NOM_CMP=DY; GROUP_NO=NO176	-0.113186924	136

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

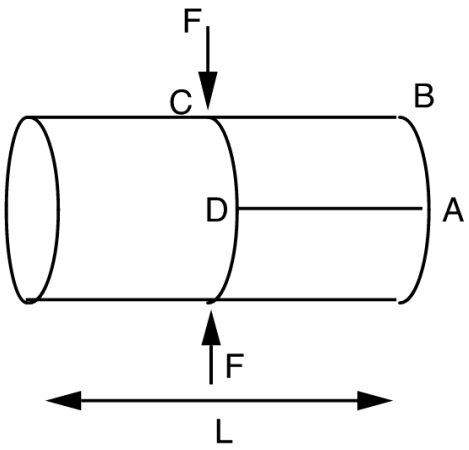
Original input: ssls20a.comm. Geometry definition: publisher diagram and original FE mesh; no separate CAD definition verified.

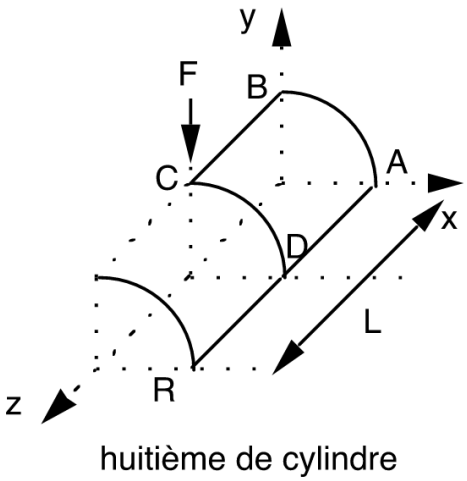
Publisher documentation: SSLS20

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Pinched cylindrical shell with a diaphragm

shells and composites





huitième de cylindre

Longueur	L = 600
Rayon	R = 300
Epaisseur	t = 3

Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_CMP=DY; GROUP_NO=C	-1.87762717e-05	76

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

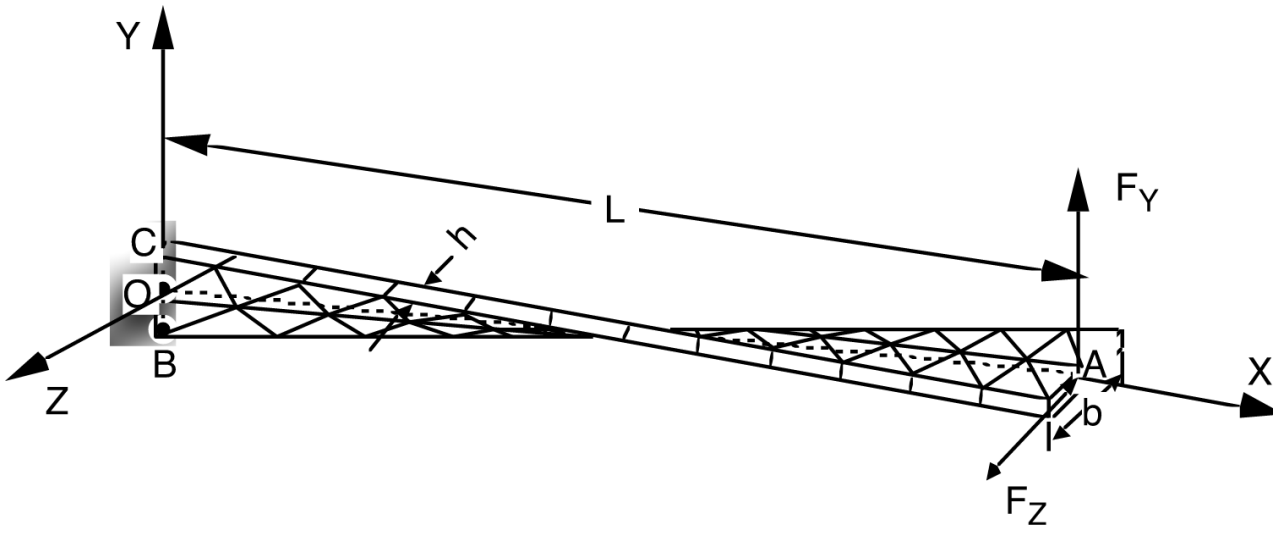
Original input: ssls104a.comm. Geometry definition: publisher diagram and original FE mesh; no separate CAD definition verified.

Publisher documentation: SSLS104

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Helicoidal shell under concentrated loads

shells and composites



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_CMP=DY; GROUP_NO=A	-0.001717466212	82
NOM_CMP=DZ; GROUP_NO=A	0.005411337178	92
NOM_CMP=DY; GROUP_NO=A	0.00175032568	102

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

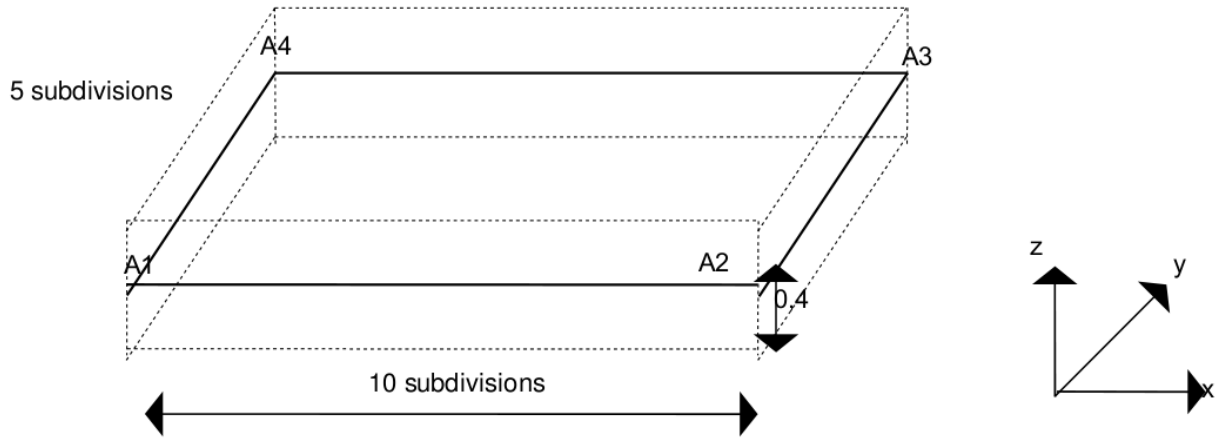
Original input: ssls108a.comm. Geometry definition: publisher diagram and original FE mesh; no separate CAD definition verified.

Publisher documentation: SSLS108

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Eccentric composite plates

shells and composites



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_CHAM=DEPL; NOM_CMP=DX; GROUP_NO=N1; NUME_ORDRE=1	-3.68041952e-06	236
NOM_CHAM=DEPL; NOM_CMP=DY; GROUP_NO=N1; NUME_ORDRE=1	-4.93941638e-07	246
NOM_CHAM=DEPL; NOM_CMP=DZ; GROUP_NO=N1; NUME_ORDRE=1	-0.00569776359	256

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

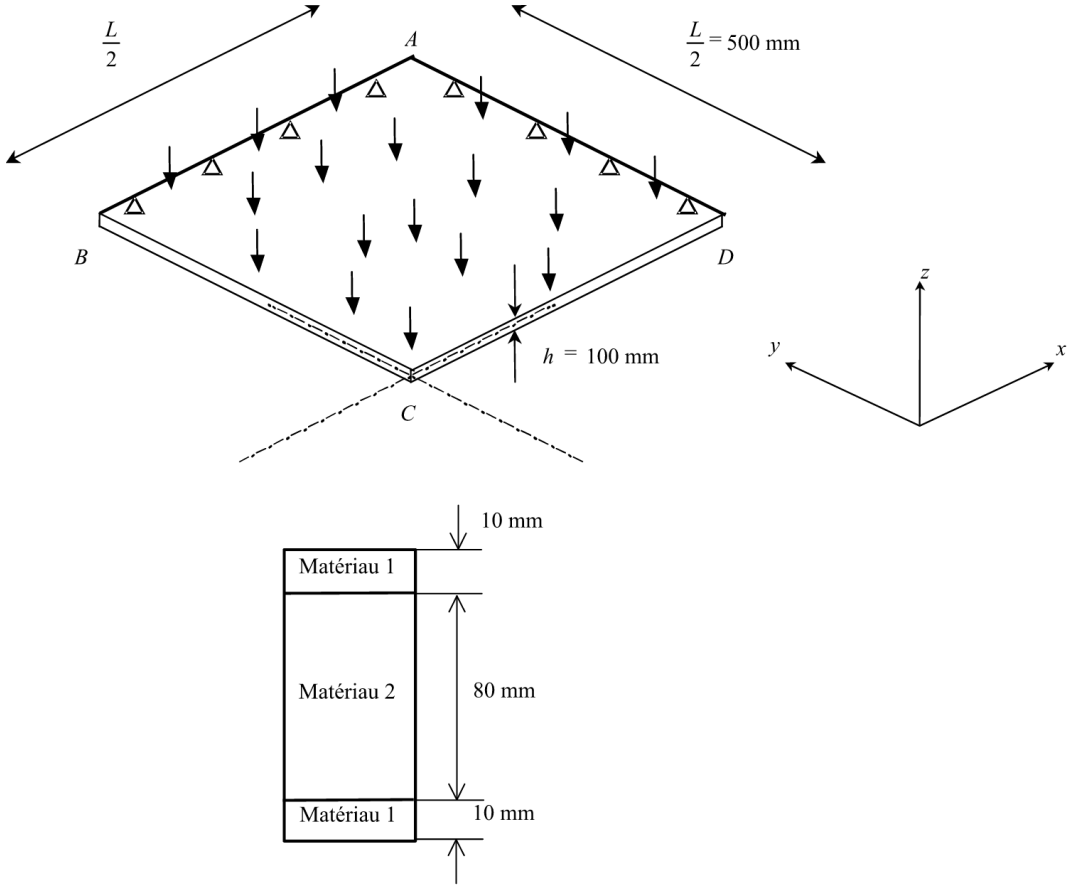
Original input: ssls112a.comm. Geometry definition: original GIBI script supplied.

Publisher documentation: SSLS112

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Composite square plate under uniform pressure

shells and composites



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_CHAM=DEPL; NOM_CMP=DZ; GROUP_NO=C; NUME_ORDRE=1	41.8305861	132
NOM_CHAM=DEPL; NOM_CMP=DY; GROUP_NO=C; NUME_ORDRE=1	0	143
NOM_CHAM=DEPL; NOM_CMP=DX; GROUP_NO=C; NUME_ORDRE=1	0	155

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

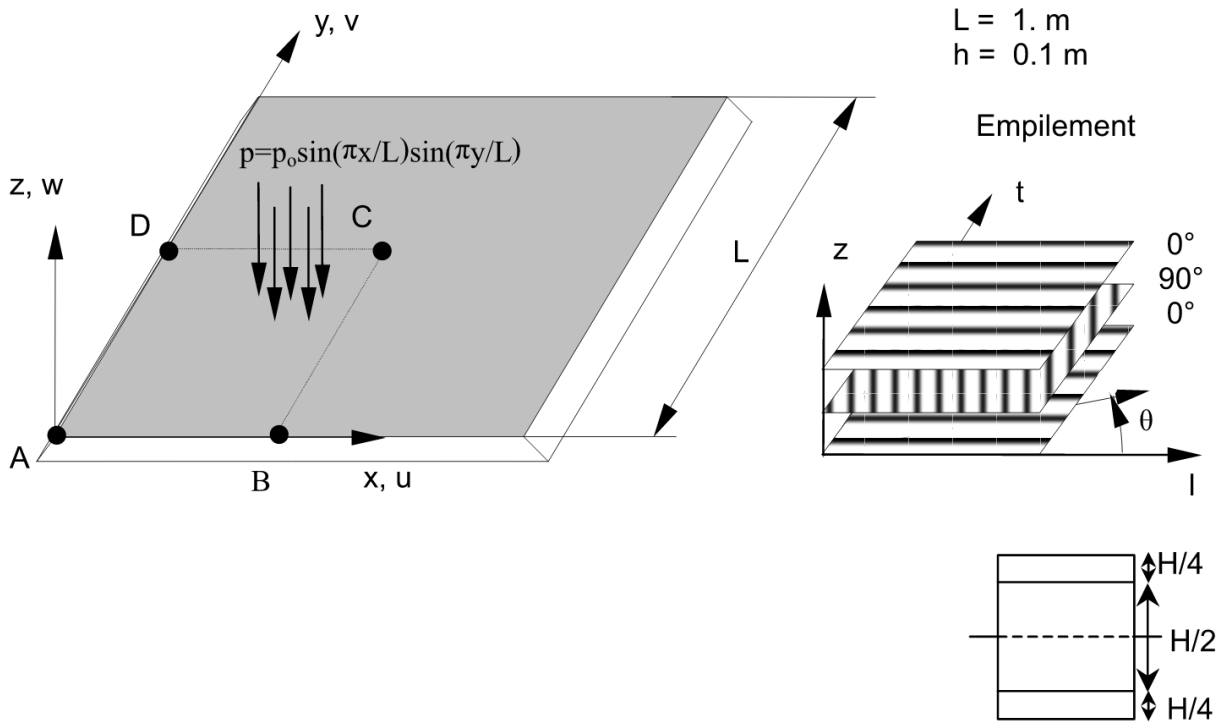
Original input: ssls115a.comm. Geometry definition: original GIBI script supplied.

Publisher documentation: SSLS115

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Three-layer composite plate under sinusoidal loading

shells and composites



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_CHAM=DEPL; NOM_CMP=DZ; GROUP_NO=C; NUME_ORDRE=1	-0.074450065	161
NOM_CMP=SIXZ; GROUP_NO=D	-0.029984866	176
NOM_CMP=SIYZ; GROUP_NO=B	-0.020984635	185

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

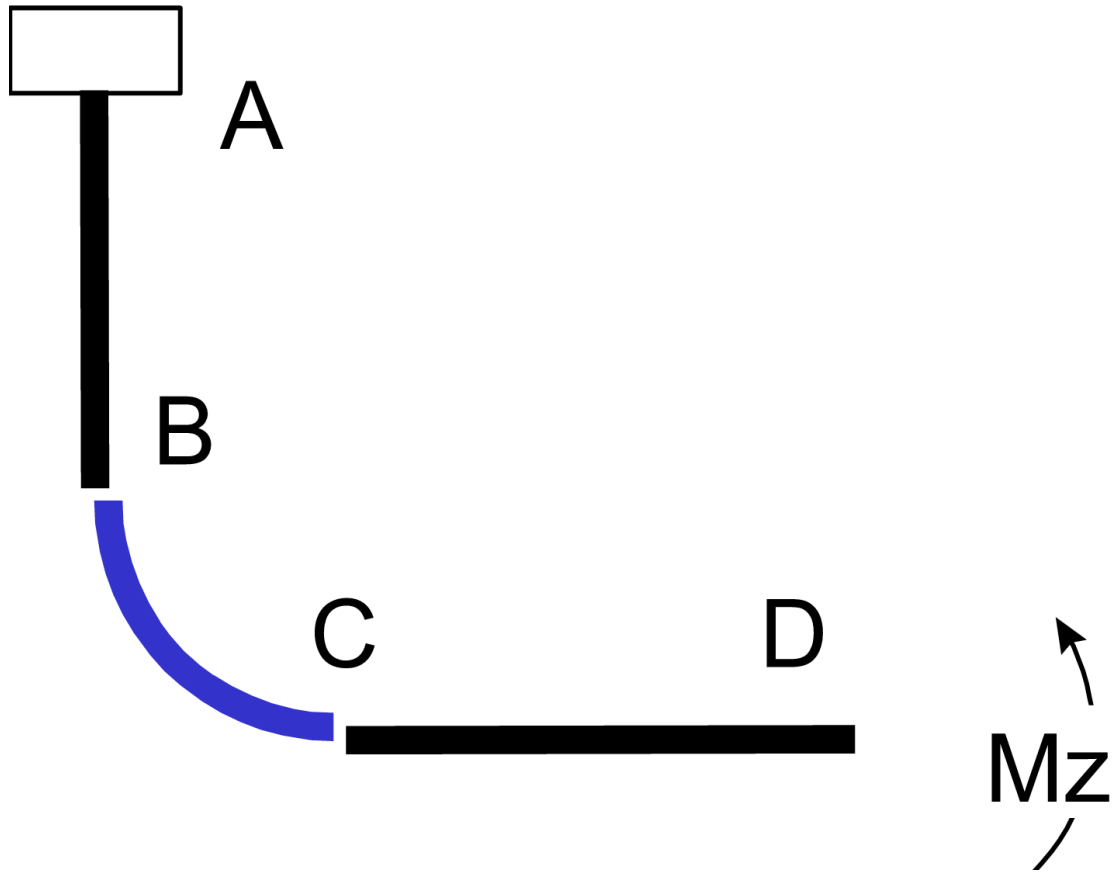
Original input: ssls504a.comm. Geometry definition: original GIBI script supplied.

Publisher documentation: SSLS504

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Bending of an elbow pipe with shell and beam models

shells and composites



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_CHAM=DEPL; NOM_CMP=DY; GROUP_NO=D; NUME_ORDRE=1	0.01942060435	118
NOM_CHAM=DEPL; NOM_CMP=DZ; GROUP_NO=D; NUME_ORDRE=1	-0.01570654759	159

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

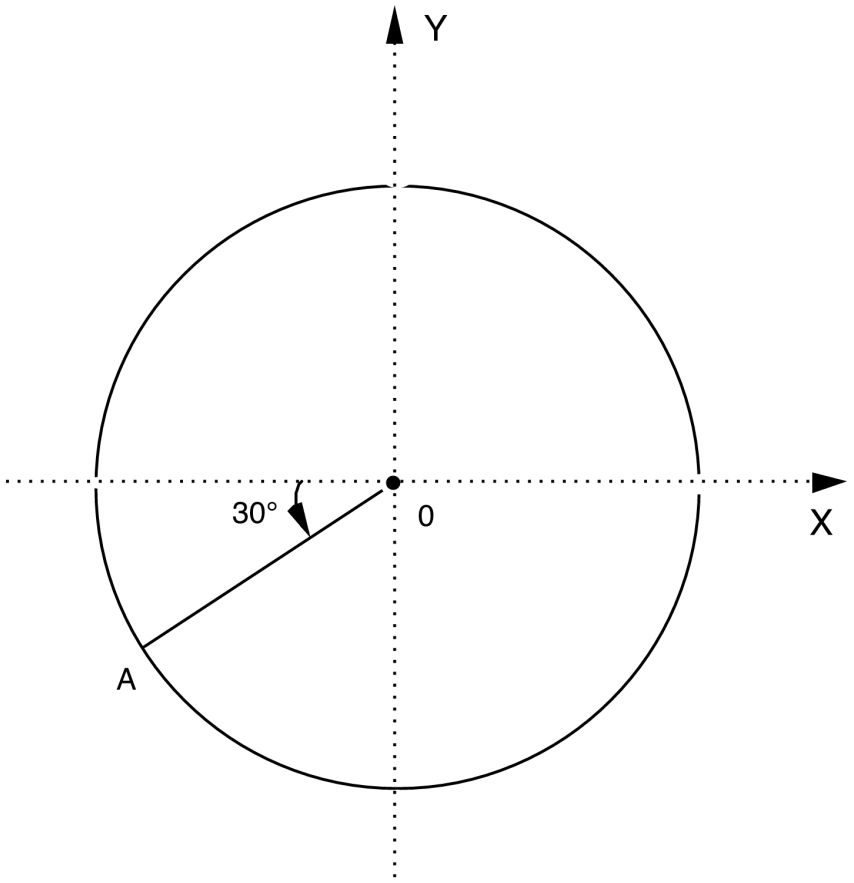
Original input: sslx102a.comm. Geometry definition: original GIBI script supplied.

Publisher documentation: SSLX102

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Mixed-mode stress intensity in a cracked circular plate

fracture and crack propagation



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_PARA=G	2.521322735e-05	209
NOM_PARA=K1	2.006703832	218
NOM_PARA=K2	0.9877740428	227

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

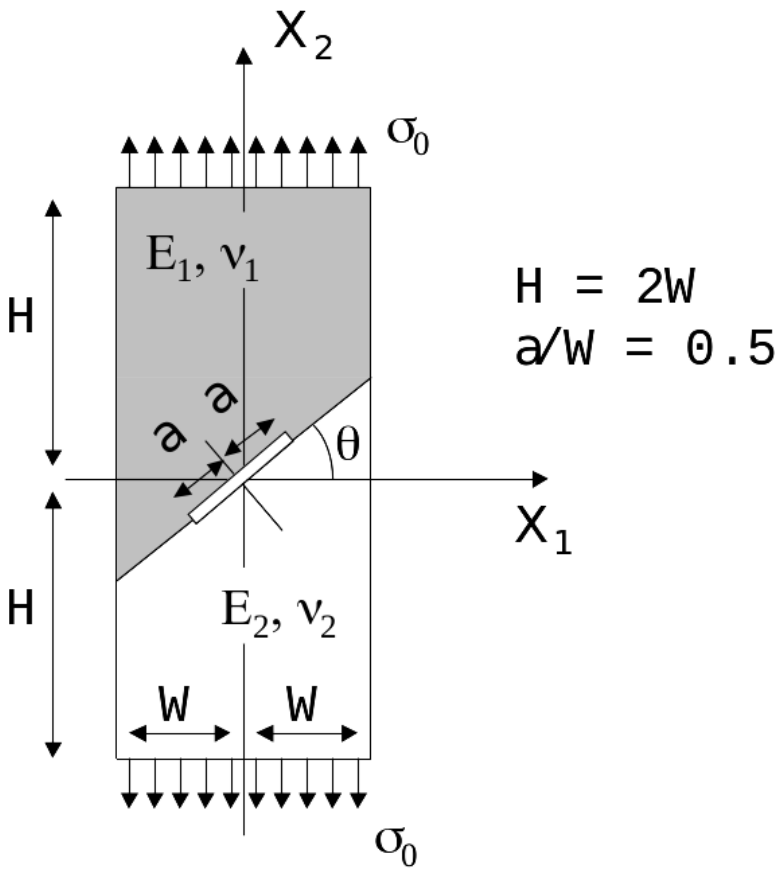
Original input: sslp103a.comm. Geometry definition: publisher diagram and original FE mesh; no separate CAD definition verified.

Publisher documentation: SSLP103

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Oblique crack in a finite bimaterial plate

fracture and crack propagation



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_PARA=G	96.76330997	87
NOM_PARA=G	100.9311937	104
NOM_PARA=G	100.9513117	122

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

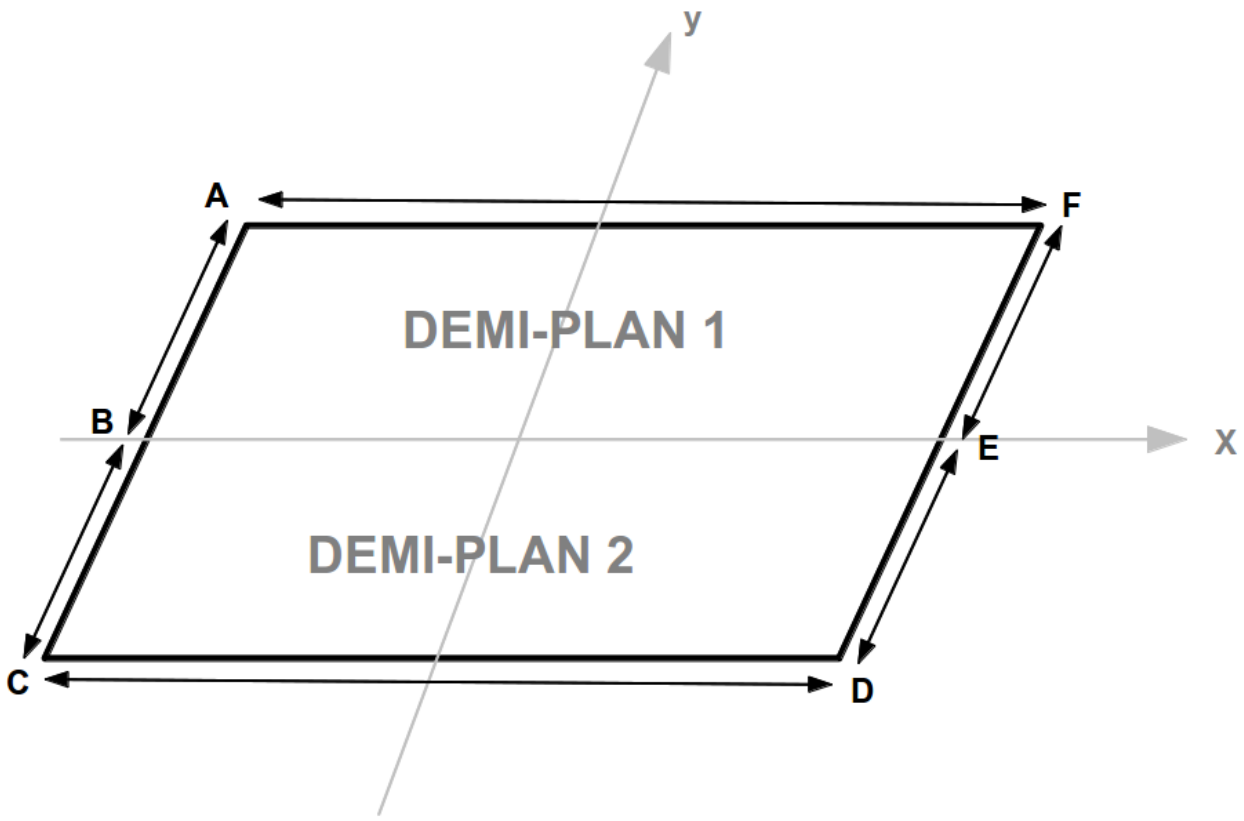
Original input: sslp311a.comm. Geometry definition: original GIBI script supplied.

Publisher documentation: SSLP311

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Deflected crack at an interface between elastic half-planes

fracture and crack propagation



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_PARA=G	86.03905533	118
NOM_PARA=G	85.04789218	127
NOM_PARA=G	85.0022051	136

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

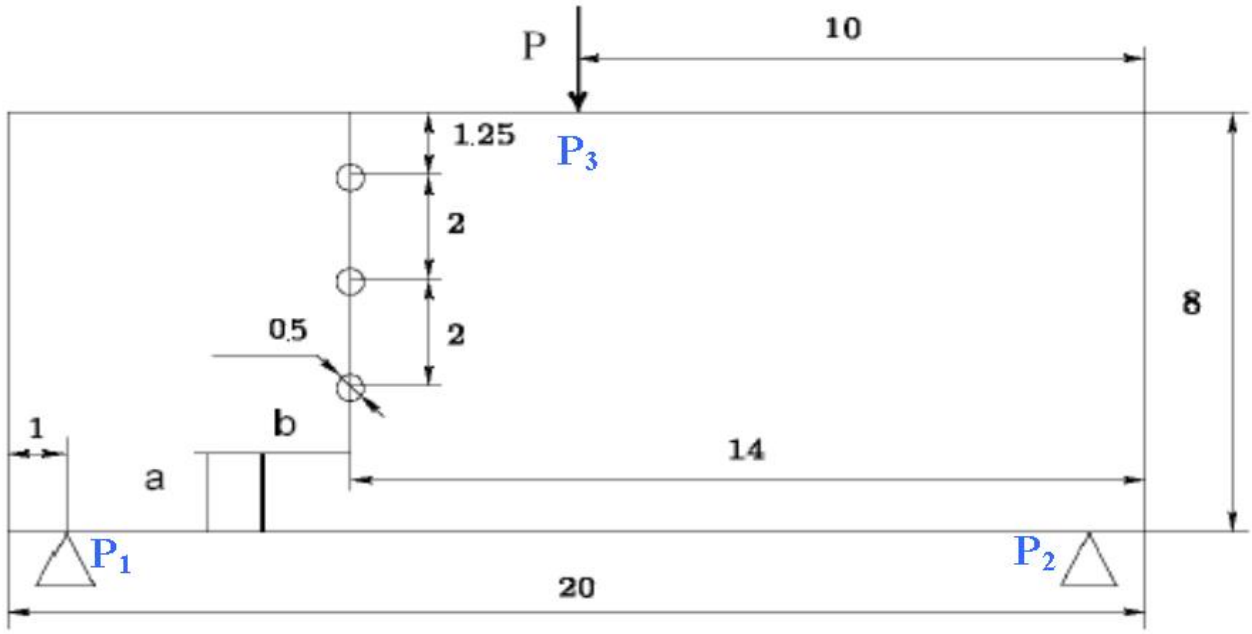
Original input: sslp314a.comm. Geometry definition: original GIBI script supplied.

Publisher documentation: SSLP314

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Crack propagation in a three-hole bending specimen

fracture and crack propagation



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_PARA=K1	1.099360048	166
NOM_PARA=K2	-0.07727282425	171

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

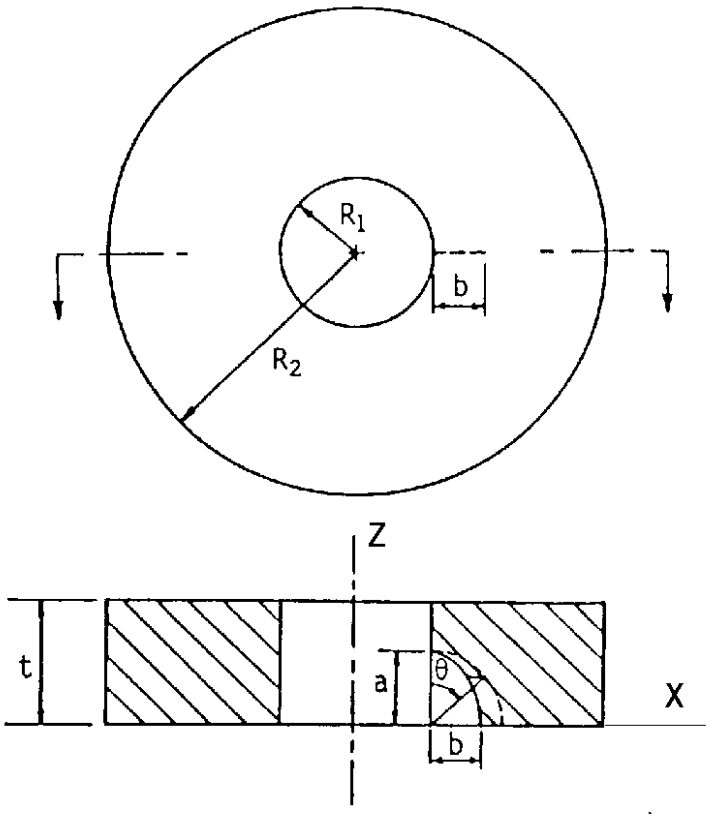
Original input: sslp322a.comm. Geometry definition: original GIBI script supplied.

Publisher documentation: SSLP322

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Quarter-elliptical corner crack in a rotating thick disc

fracture and crack propagation



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_PARA=K1	54818340.5	139
NOM_PARA=K1	74154792.24	150
NOM_PARA=K1	101830114.4	161

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

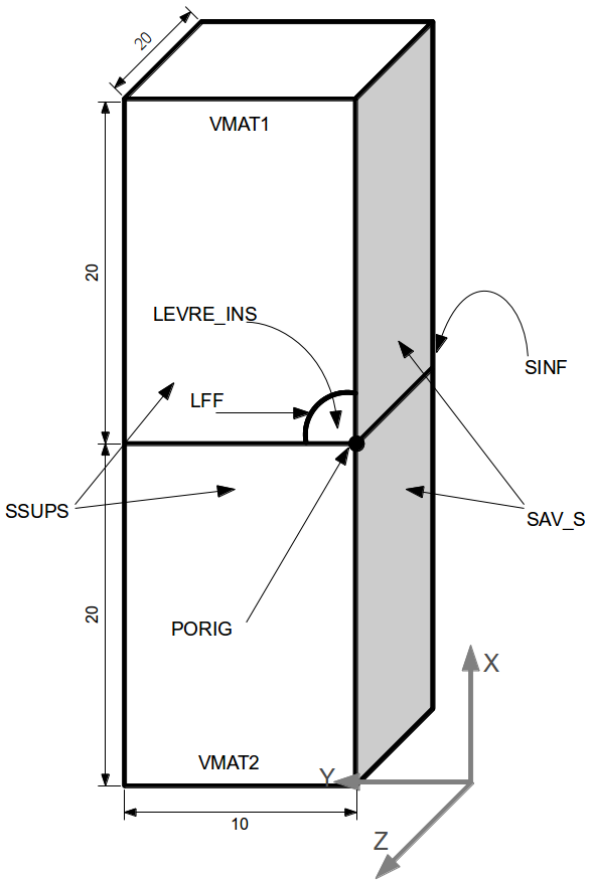
Original input: sslv311a.comm. Geometry definition: original GIBI script supplied.

Publisher documentation: SSLV311

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Semi-elliptical crack perpendicular to a bimaterial interface

fracture and crack propagation



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_PARA=G	8.826372383e-06	155
NOM_PARA=G	8.664786576e-06	165
NOM_PARA=G	8.593146496e-06	175

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

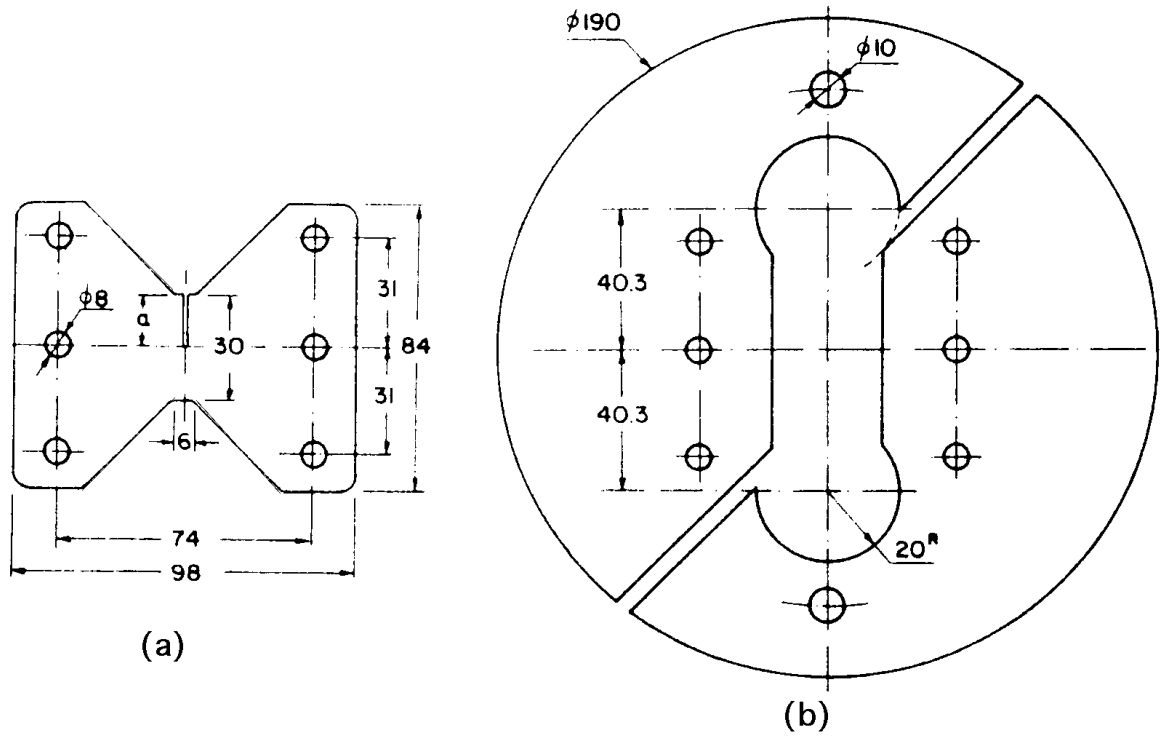
Original input: sslv312a.comm. Geometry definition: original GIBI script supplied.

Publisher documentation: SSLV312

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Mode II fracture of an elastoplastic specimen

fracture and crack propagation



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_PARA=TOTALE	331.0178901	127
NOM_PARA=TOTALE	1099.076686	135
NOM_PARA=TOTALE	1900.125179	142

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

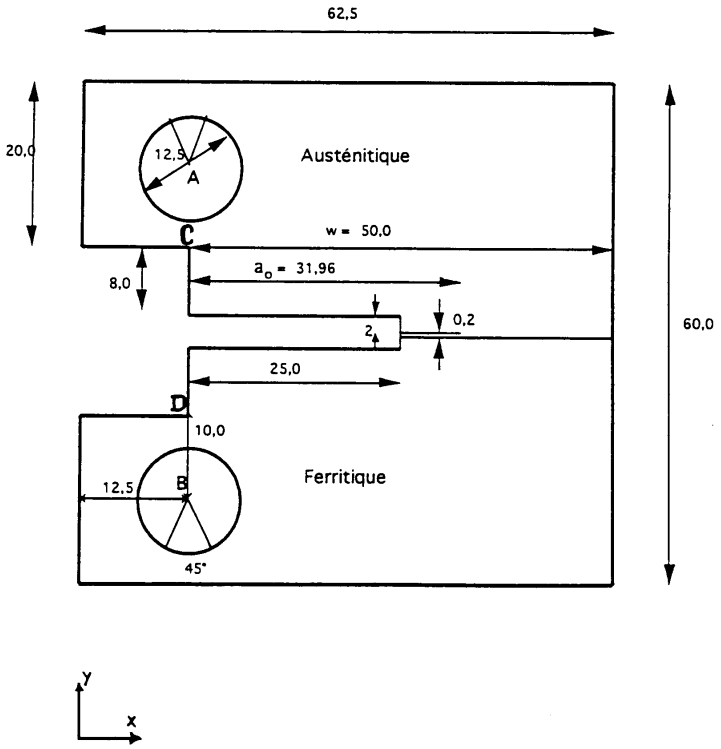
Original input: ssnp311a.comm. Geometry definition: original GIBI script supplied.

Publisher documentation: SSNP311

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Interface-parallel crack in a bimetallic CT specimen

fracture and crack propagation



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_PARA=G	6.00191581	404
NOM_PARA=G	23.23136431	413
NOM_PARA=G	48.17527731	422

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

Original input: ssnp312a.comm. Geometry definition: original GIBI script supplied.

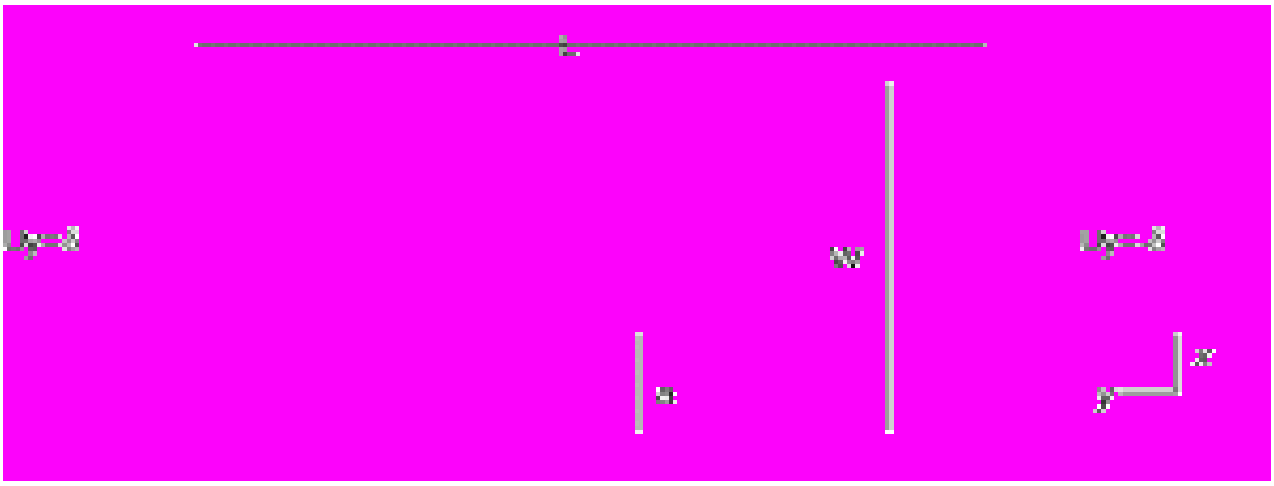
Publisher documentation: SSNP312

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

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Edge crack in an elastoplastic finite plate

fracture and crack propagation



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_PARA=DY	0.09875689897	80
NOM_PARA=G_EPSI	3.290809022	125
NOM_PARA=G_EPSI	13.6031881	132

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

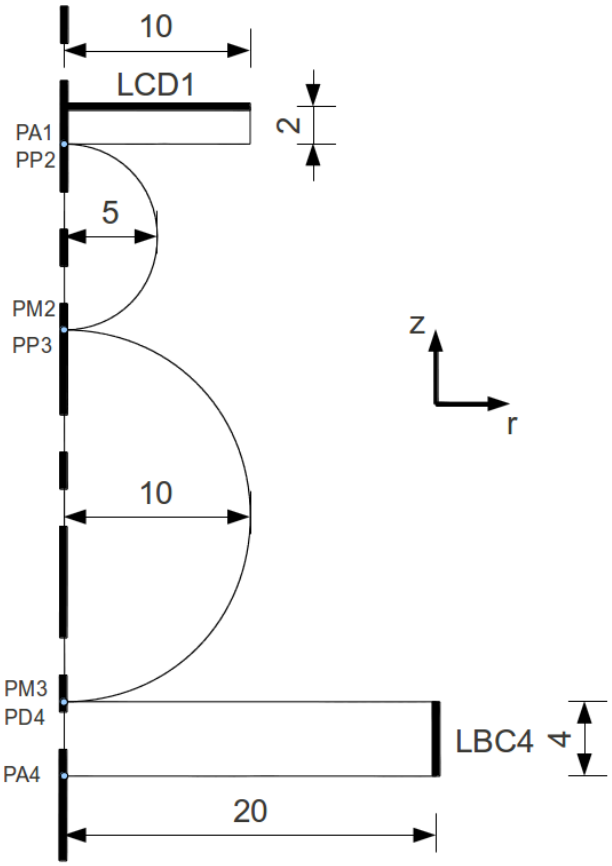
Original input: ssnp110a.comm. Geometry definition: original GIBI script supplied.

Publisher documentation: SSNP110

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Contact between multiple elastic bodies

contact and friction



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_CHAM=DEPL; NOM_CMP=DY; GROUP_NO=PA4; NUME_ORDRE=1	-0.841259654	194
NOM_PARA=DY	17496.61041	231
NOM_PARA=JEU	-1.748532651	273

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

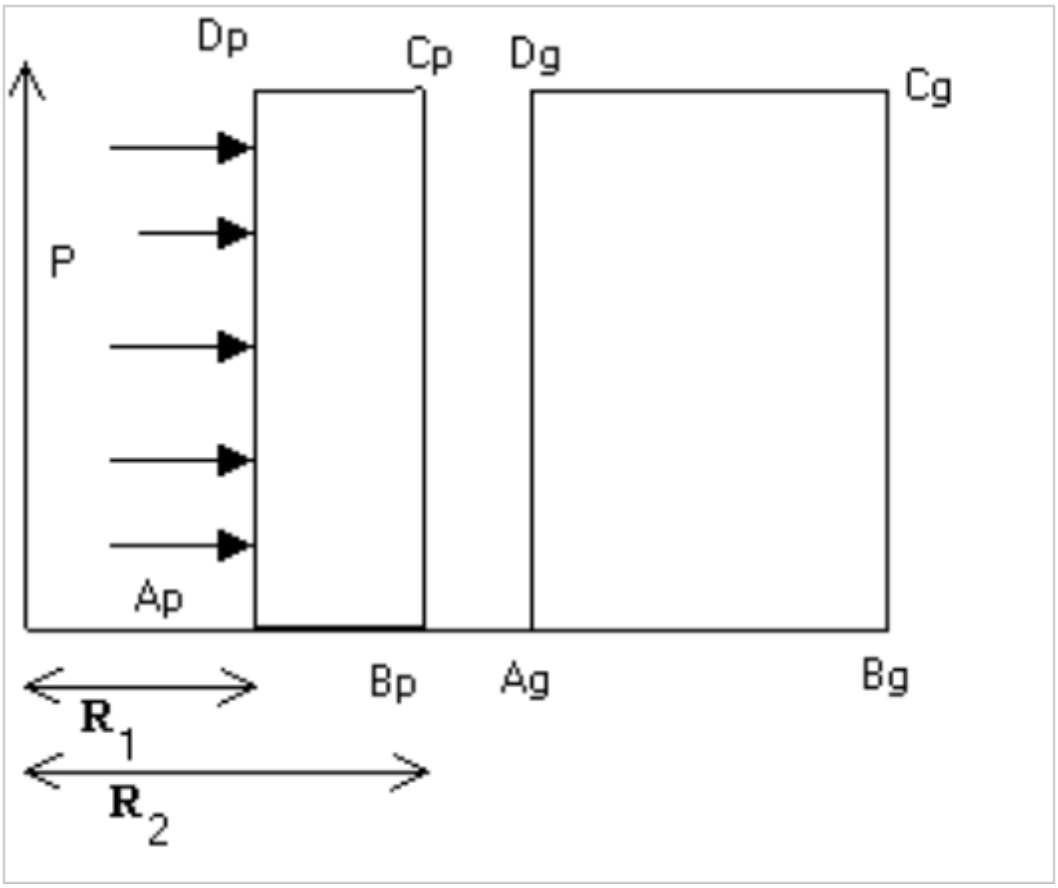
Original input: ssna102a.comm. Geometry definition: original GIBI script supplied.

Publisher documentation: SSNA102

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Viscoelastic pressurised cylinder with contact

contact and friction



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_CHAM=DEPL; NOM_CMP=DX; GROUP_NO=BG; NUME_ORDRE=125	0.00211960763	196
NOM_CHAM=SIGM_NOEU; NOM_CMP=SIXX; GROUP_NO=BG; NUME_ORDRE=125	-9.5711422e-06	207
NOM_CHAM=SIGM_NOEU; NOM_CMP=SIYY; GROUP_NO=BG; NUME_ORDRE=125	0.000269662874	218

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

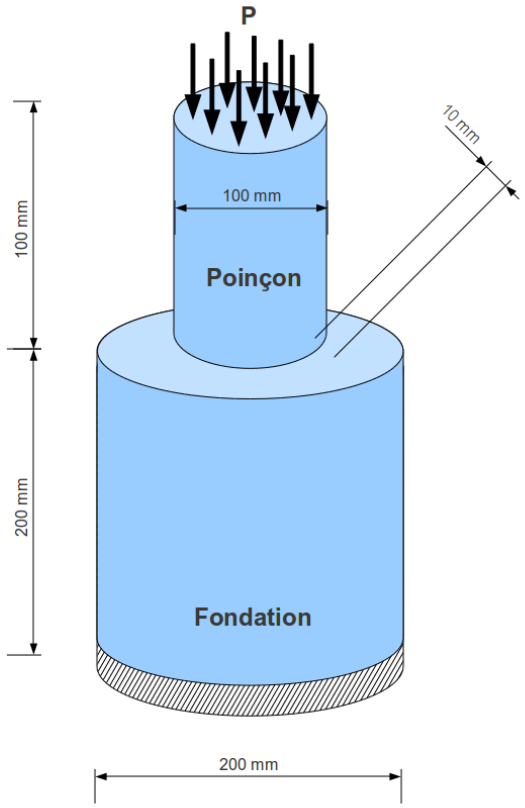
Original input: ssna105a.comm. Geometry definition: original GIBI script supplied.

Publisher documentation: SSNA105

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

NAFEMS rounded punch contact benchmark

contact and friction



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_CHAM=DEPL; NOM_CMP=DY; GROUP_NO=M; INST=1.0	-0.130734196	312
NOM_CHAM=SIEF_NOEU; NOM_CMP=SIYY; GROUP_NO=M; INST=1.0	-91.10061184	322
NOM_CHAM=DEPL; NOM_CMP=DY; GROUP_NO=M; INST=1.0	-0.128580159	337

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

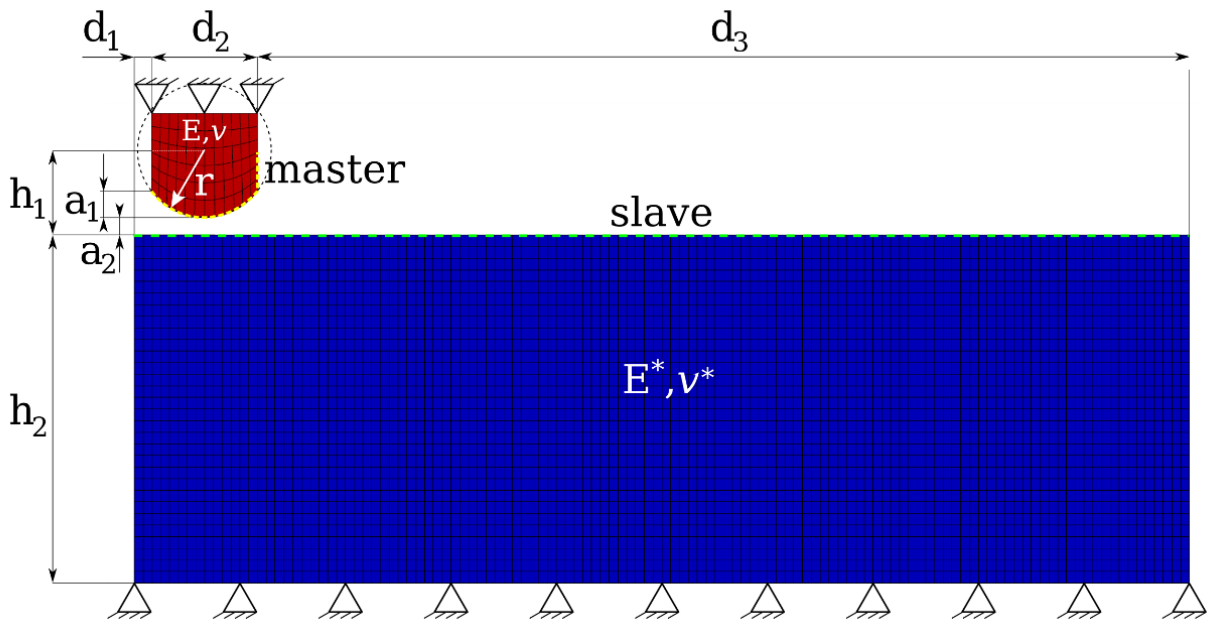
Original input: ssna122a.comm. Geometry definition: publisher diagram and original FE mesh; no separate CAD definition verified.

Publisher documentation: SSNA122

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Shallow ironing with friction and large deformation

contact and friction



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_PARA=DY	-94538776.17	159
NOM_PARA=DY	-134556789.7	163
NOM_PARA=DY	-178402887	167

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

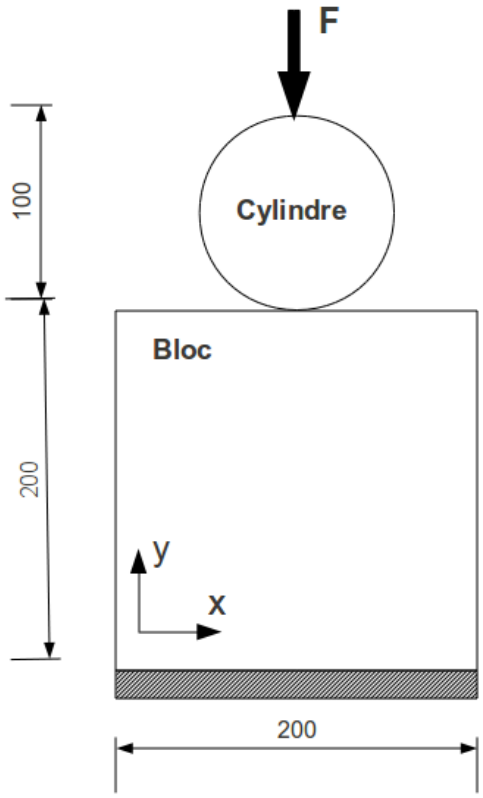
Original input: ssnp153a.comm. Geometry definition: original GIBI script supplied.

Publisher documentation: SSNP153

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

NAFEMS cylinder-roller contact benchmark

contact and friction



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_CHAM=PRES_NOEU; NOM_CMP=PRES; GROUP_NO=N5; INST=1.0	-3811.914543	139
NOM_CHAM=DEPL; NOM_CMP=DY; GROUP_NO=B; INST=1.0	-1.273028776	234
NOM_CHAM=SIEF_NOEU; NOM_CMP=SIYY; GROUP_NO=B; INST=1.0	-3806.075284	245

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

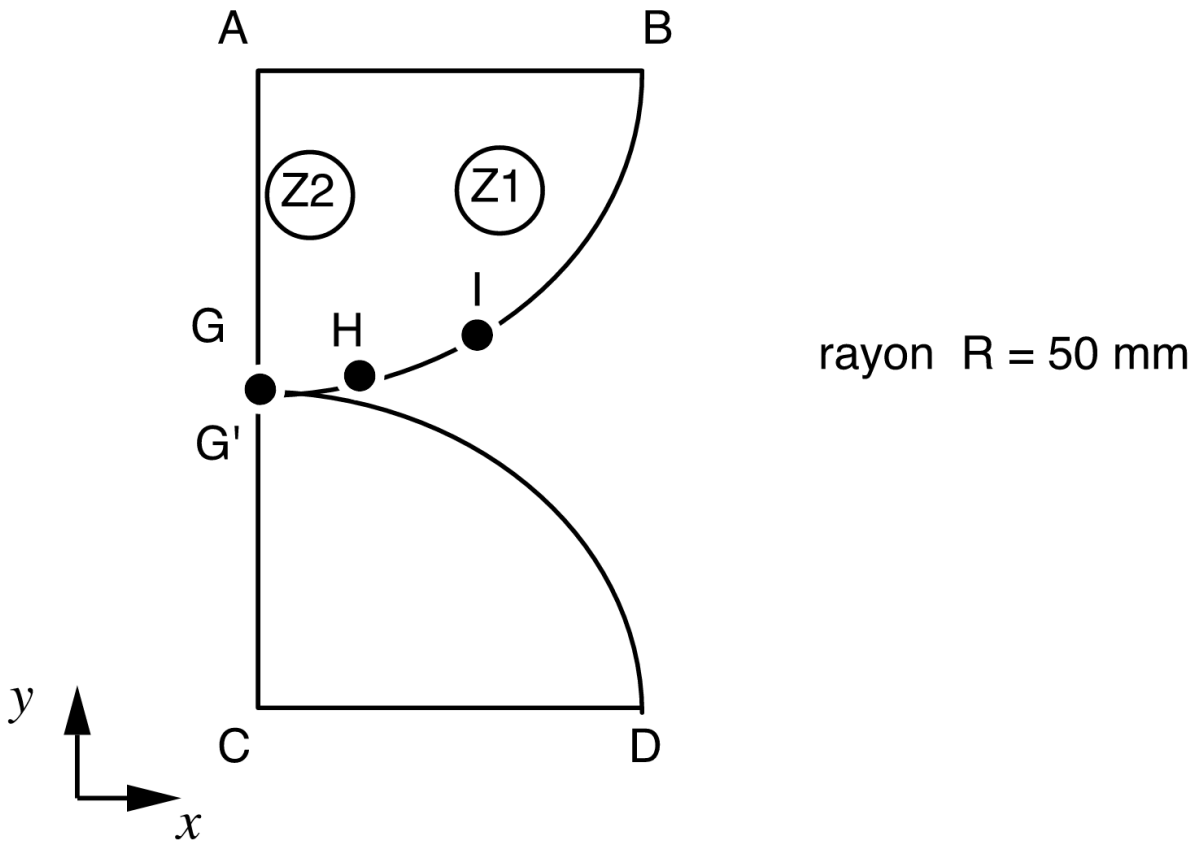
Original input: ssnp154a.comm. Geometry definition: publisher diagram and original FE mesh; no separate CAD definition verified.

Publisher documentation: SSNP154

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Hertzian contact between two spheres

contact and friction



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_CMP=SIYY; GROUP_NO=G1; GROUP_MA=M2948	-3176.335131	198
NOM_CMP=SIYY; GROUP_NO=G1; GROUP_MA=M2960	-3176.335131	229
NOM_CMP=SIYY; GROUP_NO=G1; GROUP_MA=M2972	-3176.335131	237

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

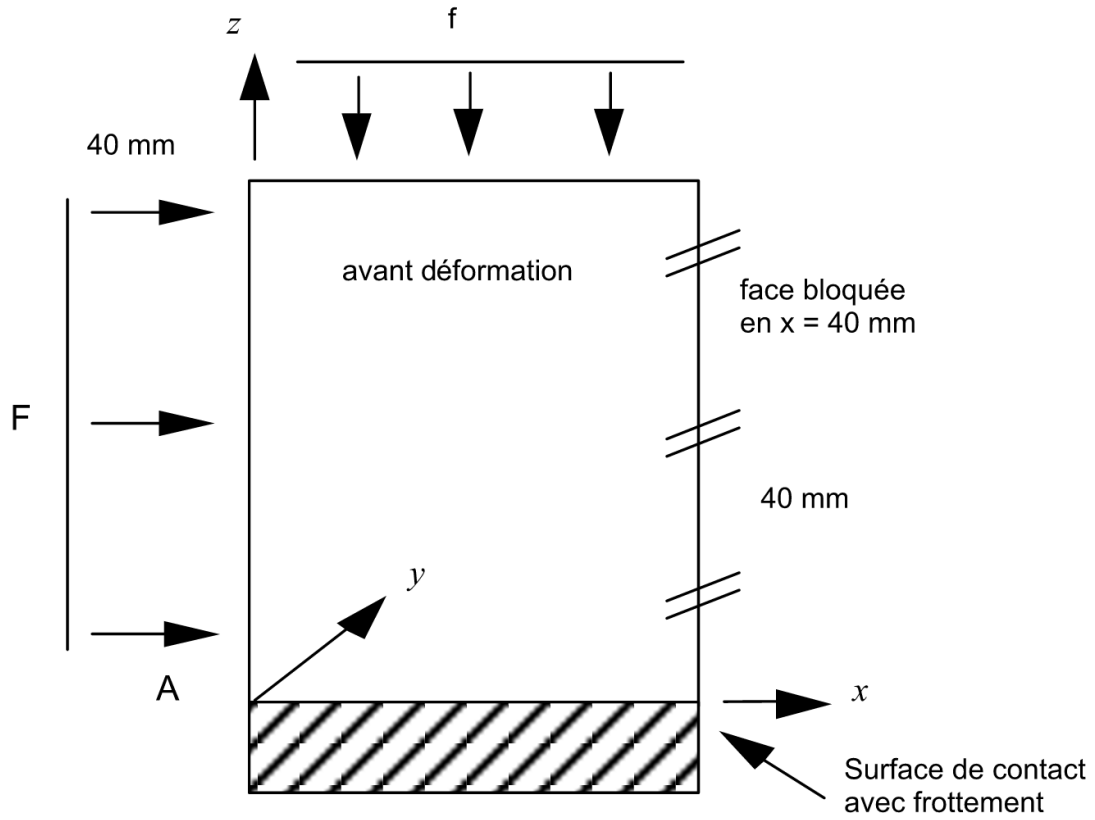
Original input: ssnv104a.comm. Geometry definition: original GIBI script supplied.

Publisher documentation: SSNV104

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Frictional contact of a plate with a rigid plane

contact and friction



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_CHAM=DEPL; NOM_CMP=DX; GROUP_NO=PPA; INST=1.0	2.864883901e-05	121
NOM_CHAM=DEPL; NOM_CMP=DX; GROUP_NO=PPB; INST=1.0	2.726866419e-05	132
NOM_CHAM=DEPL; NOM_CMP=DX; GROUP_NO=PPC; INST=1.0	2.292355773e-05	143

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

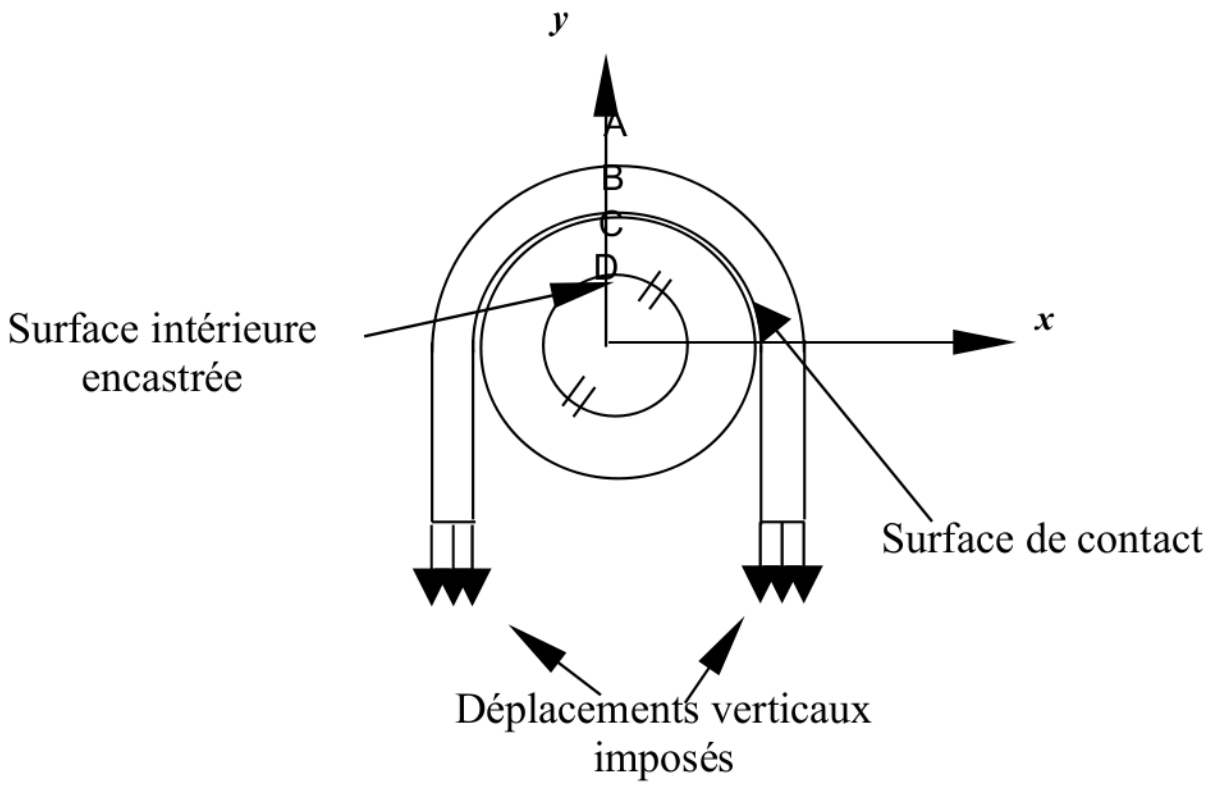
Original input: ssnv128a.comm. Geometry definition: original GIBI script supplied.

Publisher documentation: SSNV128

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Pulley-rope contact

contact and friction



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_CMP=SIYY; GROUP_NO=N37; GROUP_MA=M96	-602712.145	102
NOM_CHAM=DEPL; NOM_CMP=DX; GROUP_NO=N450; NUME_ORDRE=1	-5.64175109e-09	117
NOM_CHAM=DEPL; NOM_CMP=DX; GROUP_NO=N37; NUME_ORDRE=1	3.04866205e-09	129

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

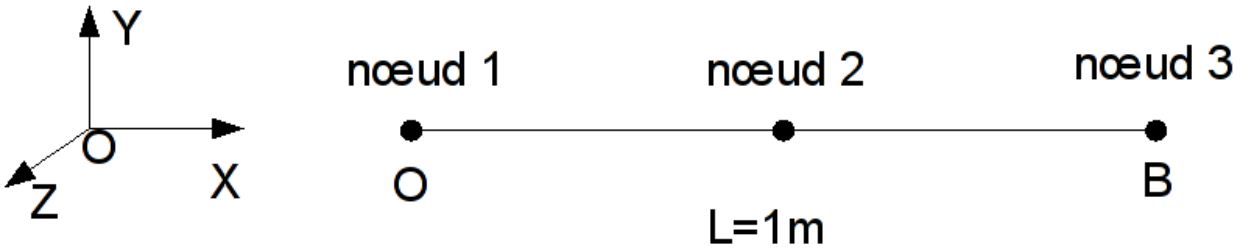
Original input: ssnv153a.comm. Geometry definition: original GIBI script supplied.

Publisher documentation: SSNV153

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Elastoplastic beam in tension and pure bending

plasticity and forming



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_CHAM=EFGE_ELNO; NOM_CMP=N; GROUP_NO=R3; GROUP_MA=SR2; NUME_ORDRE=11	3000300	148
NOM_CHAM=EFGE_ELNO; NOM_CMP=N; GROUP_NO=R3; GROUP_MA=SR2; NUME_ORDRE=21	3000600	160
NOM_CHAM=EFGE_ELNO; NOM_CMP=N; GROUP_NO=C3; GROUP_MA=SC2; NUME_ORDRE=11	4728989.41	172

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

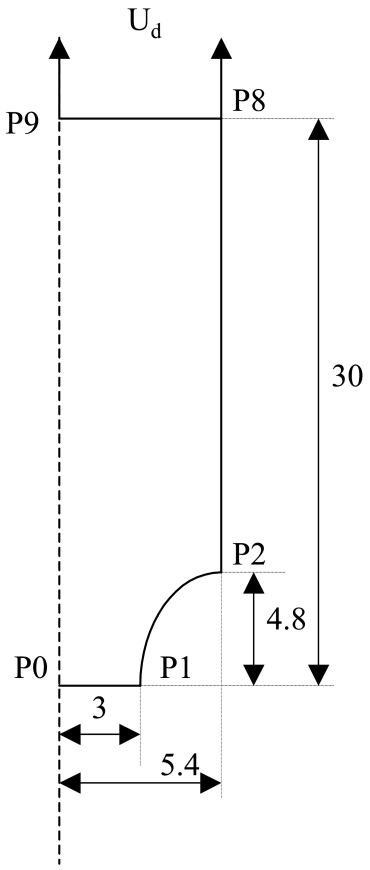
Original input: ssnl106a.comm. Geometry definition: publisher diagram and original FE mesh; no separate CAD definition verified.

Publisher documentation: SSNL106

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Large-strain notched elastoplastic specimen

plasticity and forming



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_PARA=DY	6616.59675	125
NOM_PARA=DX	-0.084493226	135
NOM_PARA=DY	21541.5707	145

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

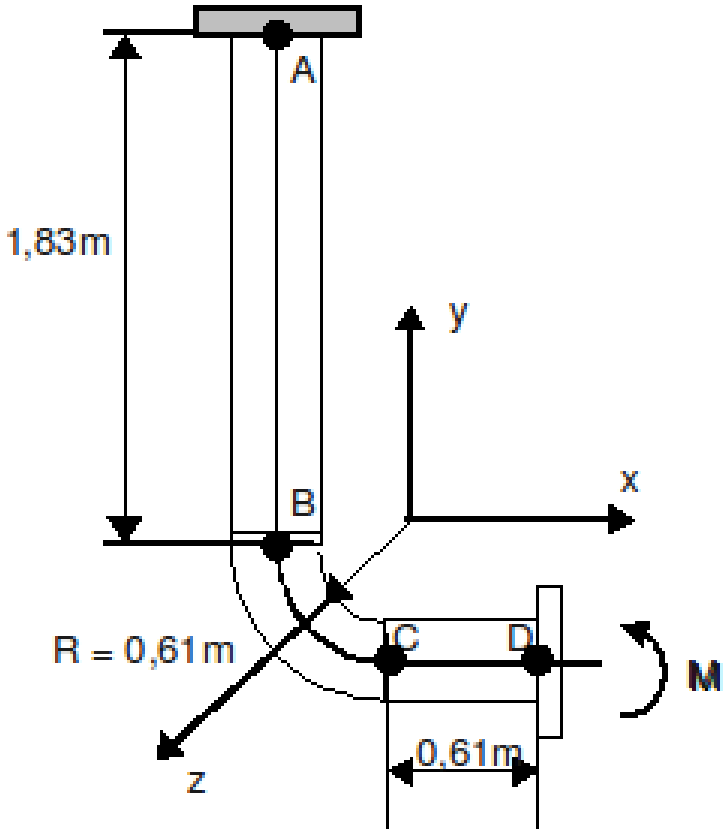
Original input: ssna303a.comm. Geometry definition: publisher diagram and original FE mesh; no separate CAD definition verified.

Publisher documentation: SSNA303

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Elastoplastic collapse of a thin elbow pipe

plasticity and forming



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NUME_ORDRE=17; PARA=ETA_PILOTAGE	1.132889676	202
NUME_ORDRE=18; PARA=ETA_PILOTAGE	1.142056698	211
NUME_ORDRE=19; PARA=ETA_PILOTAGE	1.150503493	220

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

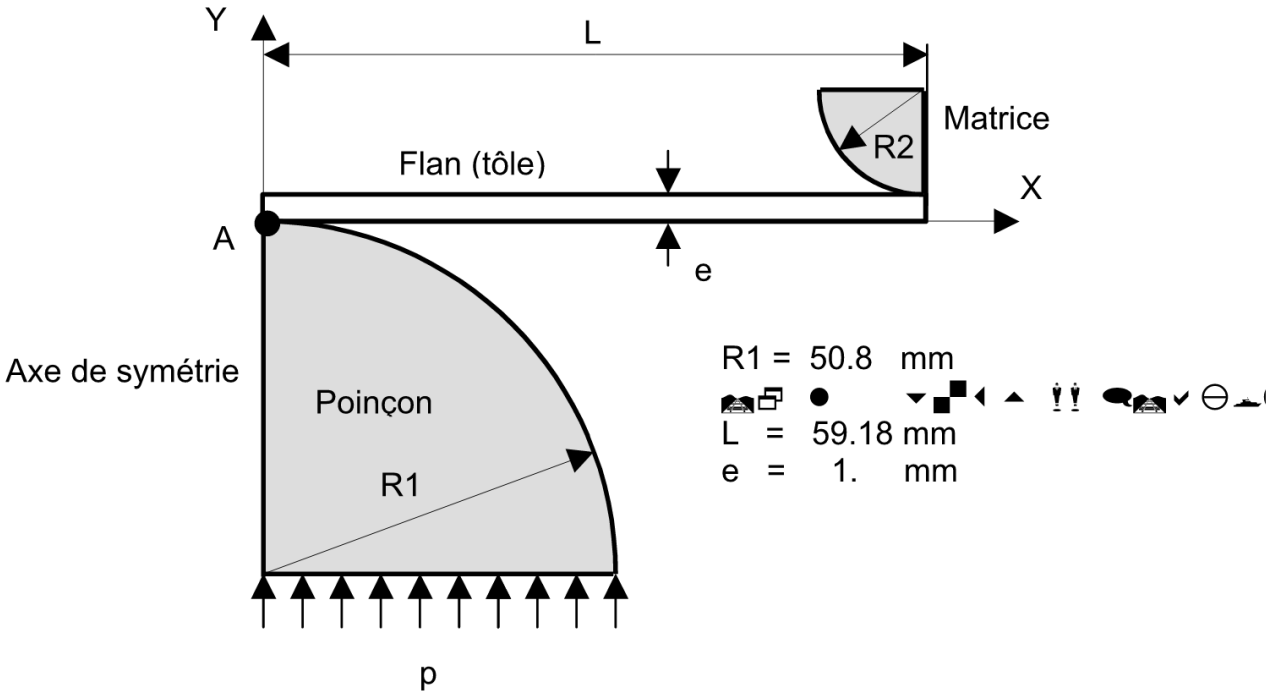
Original input: ssnl503a.comm. Geometry definition: publisher diagram and original FE mesh; no separate CAD definition verified.

Publisher documentation: SSNL503

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Sheet forming with a hemispherical punch

plasticity and forming



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_CHAM=DEPL; NOM_CMP=DY; GROUP_NO=A2; NUME_ORDRE=10	9.976750775	427
NOM_CHAM=DEPL; NOM_CMP=DY; GROUP_NO=A2; NUME_ORDRE=20	15.09052843	438
NOM_CHAM=DEPL; NOM_CMP=DY; GROUP_NO=A2; NUME_ORDRE=30	19.34401854	449

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

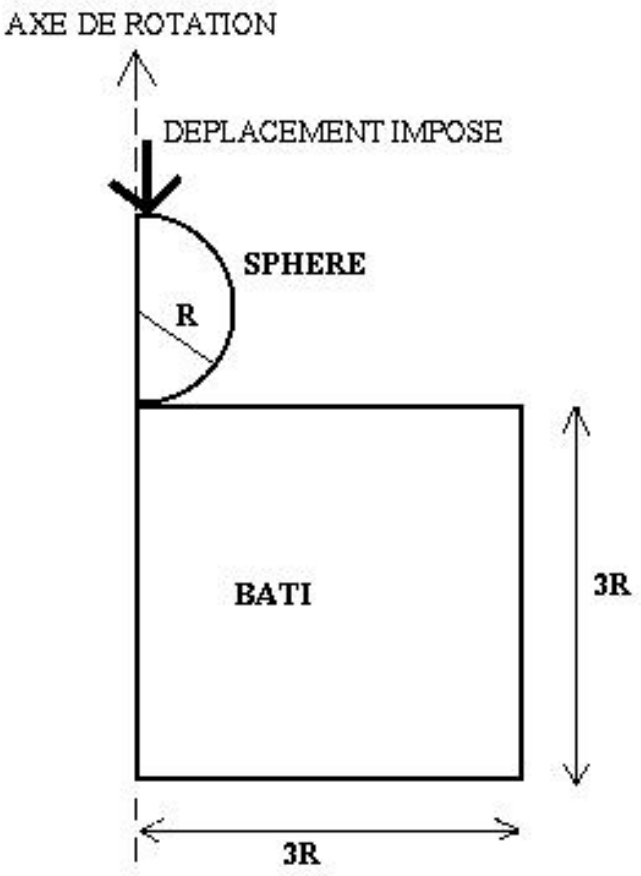
Original input: ssnv501a.comm. Geometry definition: original GIBI script supplied.

Publisher documentation: SSNV501

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Elastoplastic block indented by an elastic sphere

plasticity and forming



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_PARA=DY	-2068056.265	157
NOM_PARA=DY	-4046977.338	167
NOM_PARA=DY	-5827298.632	177

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

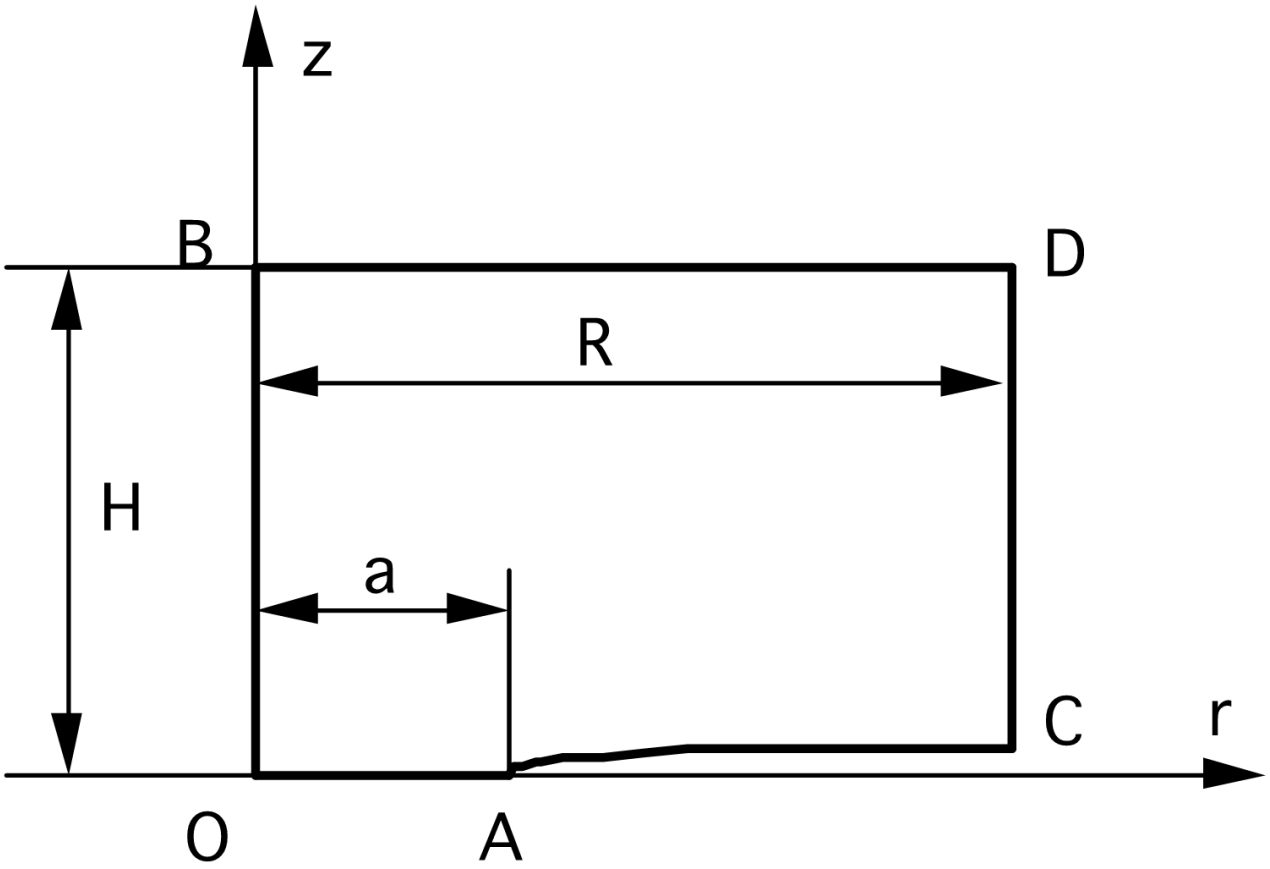
Original input: ssnv506a.comm. Geometry definition: original GIBI script supplied.

Publisher documentation: SSNV506

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Thermal shock of a radially cracked circular bar

thermomechanics



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_CHAM=TEMP; NOM_CMP=TEMP; GROUP_NO=P8; NUME_ORDRE=60	28.00131817	87
NOM_CHAM=TEMP; NOM_CMP=TEMP; GROUP_NO=P8; NUME_ORDRE=100	77.33190573	102
NOM_CHAM=TEMP; NOM_CMP=TEMP; GROUP_NO=P8; NUME_ORDRE=412	99.83407287	117

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

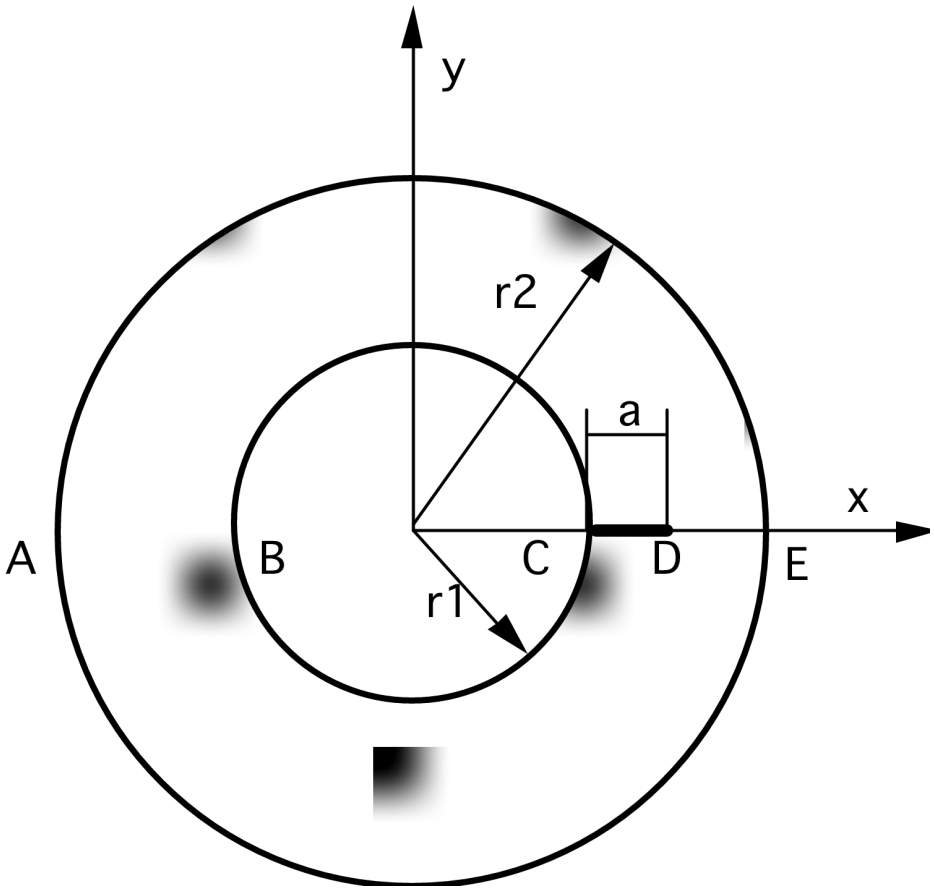
Original input: hpla310a.comm. Geometry definition: original GIBI script supplied.

Publisher documentation: HPLA310

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Cracked thick cylinder under pressure and thermal loading

thermomechanics



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_PARA=K1	1.128837239	353
NOM_PARA=K1	1.128828573	362
NOM_PARA=K1	1.128855397	371

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

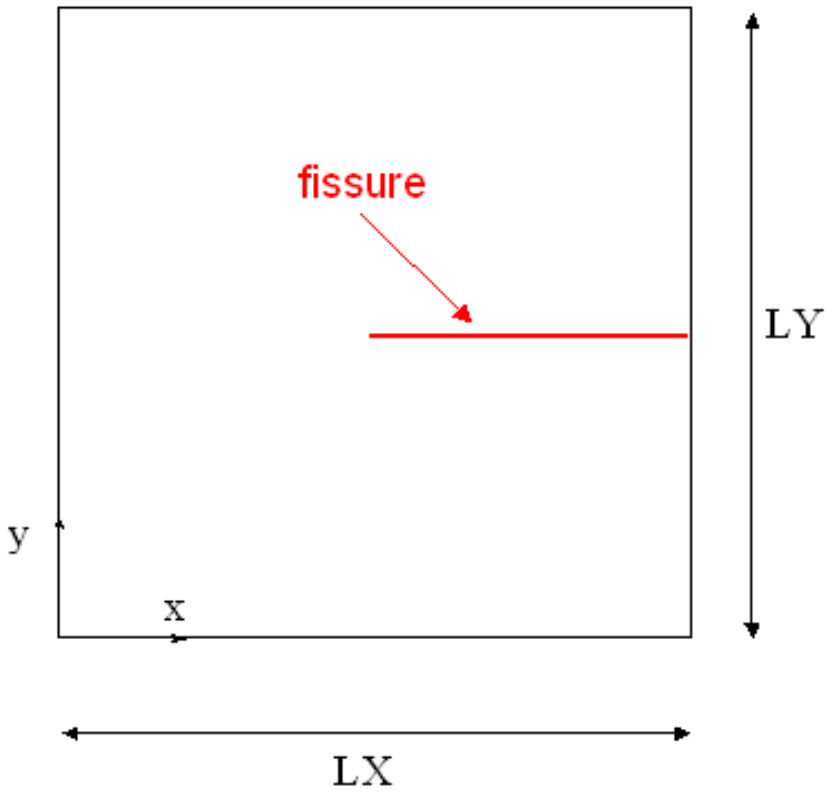
Original input: hplp310a.comm. Geometry definition: original GIBI script supplied.

Publisher documentation: HPLP310

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.

Thermoelastic cracked body with X-FEM

thermomechanics



Publisher geometry illustration; dimensions and physical setup are documented at the source link.

Published FEM results

Source output selector	FEM baseline	Source line
NOM_PARA=DX	-0.000874042629	123
NOM_PARA=DY	0.003826095851	131
NOM_PARA=K1	9032841.347	168

Values are in original source units. Full selectors and additional baselines are in the case JSON; the original command file defines the physical inputs and result extraction.

Original input: hsnv132a.comm. Geometry definition: original GIBI script supplied.

Publisher documentation: HSNV132

Source: EDF R&D, Code_Aster. Original geometry illustration reproduced with attribution. Solver assets retain their GPL notices. Admission and independent convergence review pending.